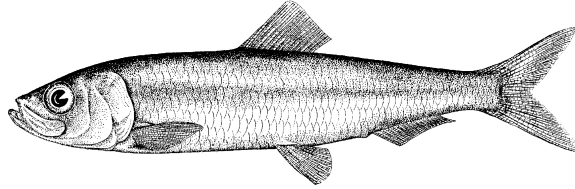


Pacific Herring preliminary data summary for Central Coast 2026

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Pacific Herring (*Clupea pallasii*). Image credit: [Fisheries and Oceans Canada](#).

Disclaimer This report contains preliminary data collected for Pacific Herring in 2026 in the Central Coast major stock assessment region (SAR). These data may differ from data used and presented in the final stock assessment.

1 Context

Pacific Herring (*Clupea pallasii*) in British Columbia are assessed as 5 major and 2 minor stock assessment regions (SARs), and data are collected and summarized on this scale (Table 1, Figure 1). Note that formal stock assessments are only done for major SARs. The Pacific Herring data collection program includes fishery-dependent and -independent data from 1951 to 2026. This includes annual time series of commercial catch data, biological samples (providing information on proportion-at-age and weight-at-age), and spawn index data conducted using a combination of surface and SCUBA surveys. In some areas, industry- and/or First Nations-operated in-season soundings programs are also conducted, and this information is used by resource managers, First Nations, and

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stakeholders to locate fish and identify areas of high and low Pacific Herring biomass to plan harvesting activities. In-season acoustic soundings are not used by stock assessment to inform the estimation of spawning biomass.

The following is a description of data collected for Pacific Herring in 2026 in the Central Coast major SAR (Figure 2). Data collected outside the SAR boundary are not included in this summary, and are not used for the purposes of stock assessment. Although we summarise data at the scale of the SAR for stock assessments, we summarise data at finer spatial scales in this report: Locations are nested within Sections, Sections are nested within Statistical Areas, and Statistical Areas are nested within SARs (Table 2). For the Central Coast major SAR, we use another level of spatial aggregation which we refer to as a ‘Group’. Note that we refer to ‘year’ instead of ‘herring season’ in this report; therefore 2026 refers to the 2025/2026 Pacific Herring season.

2 Data collection programs

The data collection program in the Central Coast reflected a number of collaborations between DFO, the Heiltsuk Nation, and the Herring Industry. Biological samples were collected by the seine test charter vessel *Ocean Venture* for 22 days from March 14th to April 4th. The primary purpose of the test charter vessel was to collect biological samples from main bodies of herring in Statistical Areas 06, 07, and 08, identified from soundings. Heiltsuk and Kitasoo Nations collected nearshore cast net herring samples again this season. Herring spawn locations were primarily identified with fixed-wing overflights. Four spawn flights were conducted this season between March 29th and April 11th. Additional spawn reconnaissance for Area 08 was conducted by the Wuikinuxv Fisheries Program.

Two dive charter vessels operated in the CC:

- The *Pachena No. 1* surveyed 15 days from April 3rd to April 17th, and
- The *Ocean Cloud* surveyed 12 days from April 2nd to April 14th.

2.1 Biological samples in Statistical Area 08

In the Central Coast major SAR, Pacific Herring in Statistical Area (SA) 08 tend to be smaller than fish in other areas. In addition, fewer biological samples are typically collected from SA 08 compared to the other areas. For example, between 1994 and 2013, on average $\sim 7\%$ of biological samples came from SA 08. In 2014 and 2015, additional resources were available to collect biological samples in SA 08, which consequently received more sampling effort than previous years, compared to the other areas. In order to avoid using non-representative biological data in the Central Coast stock assessment model, biological samples from SA 08 in 2014 and 2015 are weighted by the average historic proportion of samples in that area (i.e., $\sim 7\%$). Biological data presented in this report (e.g., proportion-at-age, weight-at-age) reflect these weights, and are considered to be representative of the Central Coast major SAR.

3 Catch and biological samples

In the 1950s and 1960s, the reduction fishery dominated Pacific Herring catch; starting in the 1970s, catch has been predominantly from roe seine and gillnet fisheries. The reduction fishery is different from current fisheries in several ways. First, the reduction fishery caught Pacific Herring of all ages, whereas current fisheries target spawning (i.e., mature) fish. Thus, reduction fisheries included an unknown number of age-1 fish which are not typically caught in current fisheries. Second, the reduction fishery has some uncertainty regarding the quantity and location of catch; however catches have been resolved to SAR and Statistical Area using fish slips as best as possible. For the roe gillnet fishery, all Pacific Herring catch has been validated by a dockside monitoring program since 1998; the catch validation program started in 1999 for the roe seine fishery. Finally, the reduction fishery operated during the summer and winter months, whereas roe fisheries typically target spawning fish between February and April.

Landed commercial catch of Pacific Herring by year and fishery is shown in Table 3 and Figure 3. Total harvested spawn-on-kelp (SOK) in 2026 in the Central Coast major SAR is shown in Table 4; we also calculate the estimated spawning biomass associated with SOK harvest. See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. Incidental mortality from fisheries and aquaculture activities is shown in Figure 4.

In 2026, 11 Pacific Herring biological samples were collected and processed for the Central Coast major SAR (Table 5, Table 6). The nearshore sampling program is a multi-year pilot study using cast nets (Figure 10, Figure 11, and Figure 12). Differences between biological data collected from two sampling protocols regarding the number-, proportion-, weight-, and length-at-age for Pacific Herring in 2026 in the Central Coast major SAR are shown in Table 7, Table 8, Table 9, and Table 10, respectively. Summaries of data collected by the nearshore sampling program are shown in Table 11, Table 12, Table 13, and Table 14. We compare length-, weight, and proportion-at-age as well as length-weight relationships for nearshore and seine test samples in Figure 13, Figure 14, Figure 15, and Figure 16, respectively. Note that only biological data from the seine samples are used for the purposes of stock assessment. The locations in which the biological samples were collected are presented in Figure 5. Included herein are biological summaries of observed proportion-, number-, weight-, and length-at-age (Figure 6, Table 15, and Figure 7, respectively). We also show the percent change in weight and length for age-3 and age-6 fish (Figure 8 & Figure 9, respectively). Some Statistical Areas tend to have larger fish at a given age (Figure 17, Table 16). Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. Only representative biological samples are included, where ‘representative’ indicates whether the Pacific Herring sample in the set accurately reflects the larger Pacific Herring school.

3.1 Nearshore genetics project

DFO initiated a nearshore genetics project in 2022 to better understand the spatial and temporal structure of Pacific Herring stocks in British Columbia. We do this by

collecting genetic and biological data from actively spawning Pacific Herring (*Clupea pallasii*). Then we collect biological data (e.g., length, weight, age, sex) and genetic data from the fish. Finally, we analyse the biological and genetic data to better understand Pacific Herring. Genetic samples were collected in the Central Coast major SAR in 2023, 2024, and 2025 (Figure 18). Results from the nearshore genetics project will be presented in a separate document when they are available.

4 Spawn survey data

Pacific Herring spawn surveys were conducted at 45 individual locations in 2026 in the Central Coast major SAR (Table 17, and Figure 19). A summary of spawn from the last decade (2016 to 2025) is shown in Figure 20. Figure 21 shows spawn start date by decade and Statistical Area. Spawn surveys are conducted to estimate the spawn length, width, number of egg layers, and substrate type, and these data are used to estimate the index of spawning biomass (i.e., the spawn index; Figure 22, Figure 23, Figure 24, Figure 25, Table 18, and Figure 26). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. In addition, spawn surveys estimate spawn depth by Statistical Area, and Section (Figure 27). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Therefore, these data do not represent model estimates of spawning biomass, and are considered the minimum observed spawning biomass derived from egg counts. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

Some Pacific Herring Sections contribute more than others to the total spawn index, and the percentage contributed by Section varies yearly (Figure 25b, Figure 28). For example, in 2026, Section 074 contributed the most to the spawn index (41%). As with Sections, some Statistical Areas contribute more than others to the total spawn index (Figure 25c, Figure 29). An animation shows the spawn index by spawn survey location from 1951 to 2026 (Figure 30).

5 First Nations observations

The following observations were contributed by representatives of First Nations communities. These observations provide context and perspectives to this data report. In some cases we make edits for clarity and brevity, but we do not change the intent or substance of responses. Note: if your observations are missing, submit them [here](#).

5.1 Heiltsuk

Guardian watchmen had difficulty catching herring samples, and comments that there were fewer herring and fewer intertidal spawns.

Herring went into Spiller channel but didn’t stay long and migrated out towards Seaforth and Dundavin. Seine test boat followed the herring, which were travelling at 4

to 6 knots from Spiller to Dundavin through Rait narrows. The captain thought that the fish from Spiller was the same fish in Stryker, Thompson, and Waskesiu.

The inlets were 2 to 3 degrees cooler than outer waters (e.g., Spiller channel was cooler than Seaforth channel), and outer areas such as Stryker, Thompson, and Waskesiu were warmer. Fish moved westward, towards warmer water.

Stock was spread out thinner, estimate of 12k tons from gillnet vessels and 14k to 17k from the herring test boat.

Fish in Spiller seemed small, and outside fish in Stryker and Thompson were bigger. Observations of clearwater spawn in Spiller lagoon (Tate) Area 8: Big spawn from Kwakume down to Illahe. Stewart inlet: good spawn, estimate of 2k to 3k tons.

Better spawning out in Stryker, Thompson, and Waskesue.

Very little herring spawn for FSC, just a few households (6) got herring roe.

Water was a different colour, more brownish.

6 General observations

The following observations were reported by DFO Resource Management staff and DFO Science staff. These observations provide additional context to this data report.

- All reconnaissance flights departed from Port Hardy in 2026.
- First spawn observed on March 29th in Area 6 (Kitasu Bay).
- Early indications of spawning activity were absent and remained low through the season. Some late spawn activity noted in Area 7 in third week of April.
- No spawning observed in Spiller Channel or in Area 8 during surveys.
- Dive boats found more spawn than was reported.

7 Tables

Table 1. Pacific Herring stock assessment regions (SARs) in British Columbia.

Name	Code	Type
Haida Gwaii	HG	Major
Prince Rupert District	PRD	Major
Central Coast	CC	Major
Strait of Georgia	SoG	Major
West Coast of Vancouver Island	WCVI	Major
Area 27	A27	Minor
Area 2 West	A2W	Minor

Table 2. Statistical Areas, Sections, and Groups for Pacific Herring in the Central Coast major stock assessment region (SAR). Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

Region	Statistical Area	Section	Group
Central Coast	06	067	06&07
Central Coast	07	070	06&07
Central Coast	07	071	06&07
Central Coast	07	072	06&07
Central Coast	07	073	06&07
Central Coast	07	074	06&07
Central Coast	07	075	06&07
Central Coast	07	076	06&07
Central Coast	07	077	06&07
Central Coast	07	078	06&07
Central Coast	07	079	06&07
Central Coast	08	085	08
Central Coast	08	086	08

Table 3. Total landed commercial catch of Pacific Herring in metric tonnes (t) by gear type in 2026 in the Central Coast major stock assessment region (SAR). Legend: ‘Other’ represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; ‘RoeSN’ represents the roe seine fishery; and ‘RoeGN’ represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Note: data may be withheld due to privacy concerns (WP).

Gear	Catch (t)
Other	0
RoeSN	0
RoeGN	0

Table 4. Total harvested Pacific Herring spawn-on-kelp (SOK) in pounds (lb), and the associated estimate of spawning biomass in metric tonnes (t) from 2016 to 2026 in the Central Coast major stock assessment region (SAR). See the [spawn index technical report](#) for calculations to convert SOK harvest to spawning biomass. Harvest does not include the 2,411 lbs of spawn on egreia in 2018. Note: data may be withheld due to privacy concerns (WP).

Year	Harvest (lb)	Spawning biomass (t)
2016	351,953	521
2017	392,747	582
2018	289,358	429
2019	356,042	528
2020	44,857	66
2021	294,269	436
2022	0	0
2023	42,022	62
2024	30,712	46
2025	44,401	66
2026	45,437	67

Table 5. Number of Pacific Herring biological samples processed from 2016 to 2026 in the Central Coast major stock assessment region (SAR). Each sample is approximately 100 fish. Note: Nearshore samples are not used in stock assessments.

Year	Number of samples			
	Commercial	Test	Nearshore	Total
2016	5	15	0	20
2017	4	13	27	44
2018	4	12	18	34
2019	2	10	28	40
2020	0	12	5	17
2021	2	13	14	29
2022	0	14	15	29
2023	2	11	11	24
2024	0	12	9	21
2025	0	19	14	33
2026	0	11	0	11

Table 6. Number and type of Pacific Herring biological samples processed in 2026 in the Central Coast major stock assessment region (SAR). Each sample is approximately 100 fish.

Type	Gear	Use	Number of samples
Test	Seine	Test fishery	11

Table 7. Observed number-at-age of Pacific Herring by sample type in 2026 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Number-at-age								
	2	3	4	5	6	7	8	9	10
Seine test	12	336	233	103	159	38	16	10	6
Total	12	336	233	103	159	38	16	10	6

Table 8. Observed proportion-at-age of Pacific Herring by sample type in 2026 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
Seine test	0.013	0.368	0.255	0.113	0.174	0.042	0.018	0.011	0.007
Total	0.013	0.368	0.255	0.113	0.174	0.042	0.018	0.011	0.007

Table 9. Observed mean weight-at-age in grams (g) of Pacific Herring by sample type in 2026 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Mean weight-at-age (g)								
	2	3	4	5	6	7	8	9	10
Seine test	45	77	83	100	110	111	116	124	140
Total	45	77	83	100	110	111	116	124	140

Table 10. Observed mean length-at-age in millimetres (mm) of Pacific Herring by sample type in 2026 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Sample type	Mean length-at-age (mm)								
	2	3	4	5	6	7	8	9	10
Seine test	138	168	172	183	188	187	192	196	200
Total	138	168	172	183	188	187	192	196	200

Table 11. Observed number-at-age of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Number-at-age								
	2	3	4	5	6	7	8	9	10
2017	97	279	208	1111	174	454	85	74	18
2018	24	328	252	231	521	135	87	33	10
2019	15	1543	605	133	129	100	26	7	3
2020	31	10	301	66	29	23	4	4	1
2021	9	226	51	833	134	28	20	8	1
2022	51	177	308	43	527	68	19	6	3
2023	27	341	103	84	31	98	15	1	1
2024	8	73	342	45	43	26	24	2	0
2025	82	325	181	351	62	40	24	22	3

Table 12. Observed proportion-at-age of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
2017	0.039	0.112	0.083	0.444	0.070	0.182	0.034	0.030	0.007
2018	0.015	0.202	0.155	0.143	0.321	0.083	0.054	0.020	0.006
2019	0.006	0.602	0.236	0.052	0.050	0.039	0.010	0.003	0.001
2020	0.066	0.021	0.642	0.141	0.062	0.049	0.009	0.009	0.002
2021	0.007	0.173	0.039	0.636	0.102	0.021	0.015	0.006	0.001
2022	0.042	0.147	0.256	0.036	0.438	0.057	0.016	0.005	0.002
2023	0.039	0.486	0.147	0.120	0.044	0.140	0.021	0.001	0.001
2024	0.014	0.130	0.607	0.080	0.076	0.046	0.043	0.004	0.000
2025	0.075	0.298	0.166	0.322	0.057	0.037	0.022	0.020	0.003

Table 13. Observed mean weight-at-age in grams (g) of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Mean weight-at-age (g)								
	2	3	4	5	6	7	8	9	10
2017	49	69	83	93	98	115	116	120	128
2018	48	68	78	94	96	104	107	107	109
2019	52	61	67	84	97	103	102	119	114
2020	34	59	73	78	89	88	102	89	93
2021	38	58	74	82	91	94	103	120	104
2022	46	60	72	92	94	103	108	121	108
2023	36	62	70	77	101	102	109	111	107
2024	43	67	82	87	102	103	103	106	NA
2025	48	63	76	93	103	103	110	125	106

Table 14. Observed mean length-at-age in millimetres (mm) of Pacific Herring for nearshore samples from 2017 to 2025 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

Year	Mean length-at-age (mm)								
	2	3	4	5	6	7	8	9	10
2017	158	175	187	193	198	207	208	209	213
2018	154	172	178	191	193	199	201	204	200
2019	153	162	167	180	190	195	196	200	206
2020	136	162	173	178	186	184	193	187	194
2021	141	161	175	180	185	192	196	200	195
2022	150	164	176	190	193	198	201	210	203
2023	140	162	170	174	188	190	191	201	194
2024	150	171	182	186	192	198	199	194	NA
2025	151	164	172	185	191	190	193	200	193

Table 15. Observed proportion-at-age for Pacific Herring from 2016 to 2026 in the Central Coast major stock assessment region (SAR). The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

Year	Proportion-at-age								
	2	3	4	5	6	7	8	9	10
2016	0.040	0.091	0.549	0.094	0.146	0.028	0.039	0.011	0.003
2017	0.029	0.170	0.132	0.446	0.089	0.092	0.024	0.012	0.007
2018	0.090	0.168	0.174	0.147	0.295	0.066	0.041	0.014	0.005
2019	0.012	0.616	0.178	0.053	0.053	0.059	0.024	0.002	0.002
2020	0.028	0.028	0.739	0.101	0.045	0.029	0.021	0.007	0.002
2021	0.012	0.295	0.031	0.521	0.085	0.033	0.015	0.007	0.001
2022	0.105	0.163	0.376	0.045	0.243	0.047	0.013	0.006	0.002
2023	0.030	0.614	0.118	0.104	0.027	0.089	0.014	0.003	0.000
2024	0.020	0.175	0.549	0.082	0.082	0.032	0.045	0.010	0.005
2025	0.052	0.332	0.184	0.300	0.049	0.053	0.015	0.011	0.003
2026	0.013	0.368	0.255	0.113	0.174	0.042	0.018	0.011	0.007

Table 16. Sample size for Pacific Herring weight-at-age analysis in 2026 in the Central Coast major stock assessment region (SAR) by Group from the most recent decade (2017 to 2026), and the previous decade (2007 to 2016), as displayed in Figure 17. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

Group	Age	Sample size	
		Previous decade	Recent decade
06&07	2	737	435
06&07	3	5,797	2,884
06&07	4	5,567	2,803
06&07	5	2,906	2,436
06&07	6	1,545	1,276
06&07	7	659	617
06&07	8	289	252
06&07	9	87	95
06&07	10	30	39
08	2	272	69
08	3	1,031	556
08	4	923	370
08	5	304	168
08	6	167	117
08	7	53	58
08	8	14	24
08	9	3	10
08	10	2	3

Table 17. Pacific Herring spawn survey locations, start date, and spawn index in metric tonnes (t) in 2026 in the Central Coast major stock assessment region (SAR). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (NAs).

Statistical Area	Section	Location name	Start date	Spawn index (t)
06	067	Abrams Is	March 29	740
06	067	Kwakwa Cr	March 29	992
06	067	Larkin Pt	April 03	3,899
06	067	Marvin Is	March 29	887
06	067	Osment Inlt	March 29	1,345
06	067	Parsons Anch	March 29	1,264
06	067	Wilby Pt	March 29	663
07	072	Balagny Pass	March 29	605
07	072	Bird Pt	April 01	19
07	072	Boat Inlt	April 01	0
07	072	Branks It	March 29	88
07	072	Cameron Pt	March 29	1
07	072	Ivory Is	March 29	334
07	072	Lambard Inlt	April 01	26
07	072	Leighton Is	April 01	9
07	072	Moss Pass	March 29	77
07	072	Oliver Cv	April 01	8
07	072	Port Blackney	April 01	120
07	072	Powell Anch	March 29	306
07	072	Reid Pass	March 29	320
07	072	Watch Is	March 29	1,408
07	074	Ardmillan Bay	March 31	29
07	074	Cape Mark	March 31	458
07	074	Cape Swaine	March 29	1,254
07	074	Cree Pt	March 31	144
07	074	Dryad Pt	March 31	0
07	074	Edwards Pt	March 20	4,774
07	074	Houghton Is	March 31	1,519
07	074	Idol Pt	March 31	276
07	074	Kynumpt Hrbr	March 31	368
07	074	Little Thompson Bay	March 20	57
07	074	Lockhart Bay	March 11	3
07	074	Norman Morrison Bay	April 21	NA
07	074	Odin Cv	March 31	110
07	074	Ormidale Hrbr	March 31	162
07	074	Potts Is	March 31	398
07	074	Princess Alice Is	March 31	673
07	074	Stryker Is	March 31	2,146

Table 17 continued					
Statistical Area	Section	Location name	Start date	Spawn index (t)	
	07	076	Stewart Inlt	March 29	668
	07	077	E Higgins Pass	April 03	2,884
	07	078	Mathieson Chnl	April 04	21
	08	085	Illahie Inlt	April 03	468
	08	085	Kwakume Inlt	April 05	523
	08	085	Kwakume Pt	April 03	354
	08	085	Whidbey Pt	April 05	50

Table 18. Summary of Pacific Herring spawn survey data from 2016 to 2026 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Units: metres (m), and metric tonnes (t).

Year	Total length (m)	Mean width (m)	Mean number of egg layers	Spawn index (t)
2016	164,575	40	1.6	32,507
2017	125,525	39	1.5	23,517
2018	98,149	25	1.1	12,263
2019	181,750	37	1.8	46,254
2020	160,625	39	2.0	42,713
2021	128,625	41	1.5	28,674
2022	153,675	42	1.3	22,711
2023	117,250	45	1.4	17,550
2024	137,400	40	1.4	26,802
2025	134,375	41	1.4	32,503
2026	172,475	44	1.4	30,470

8 Figures

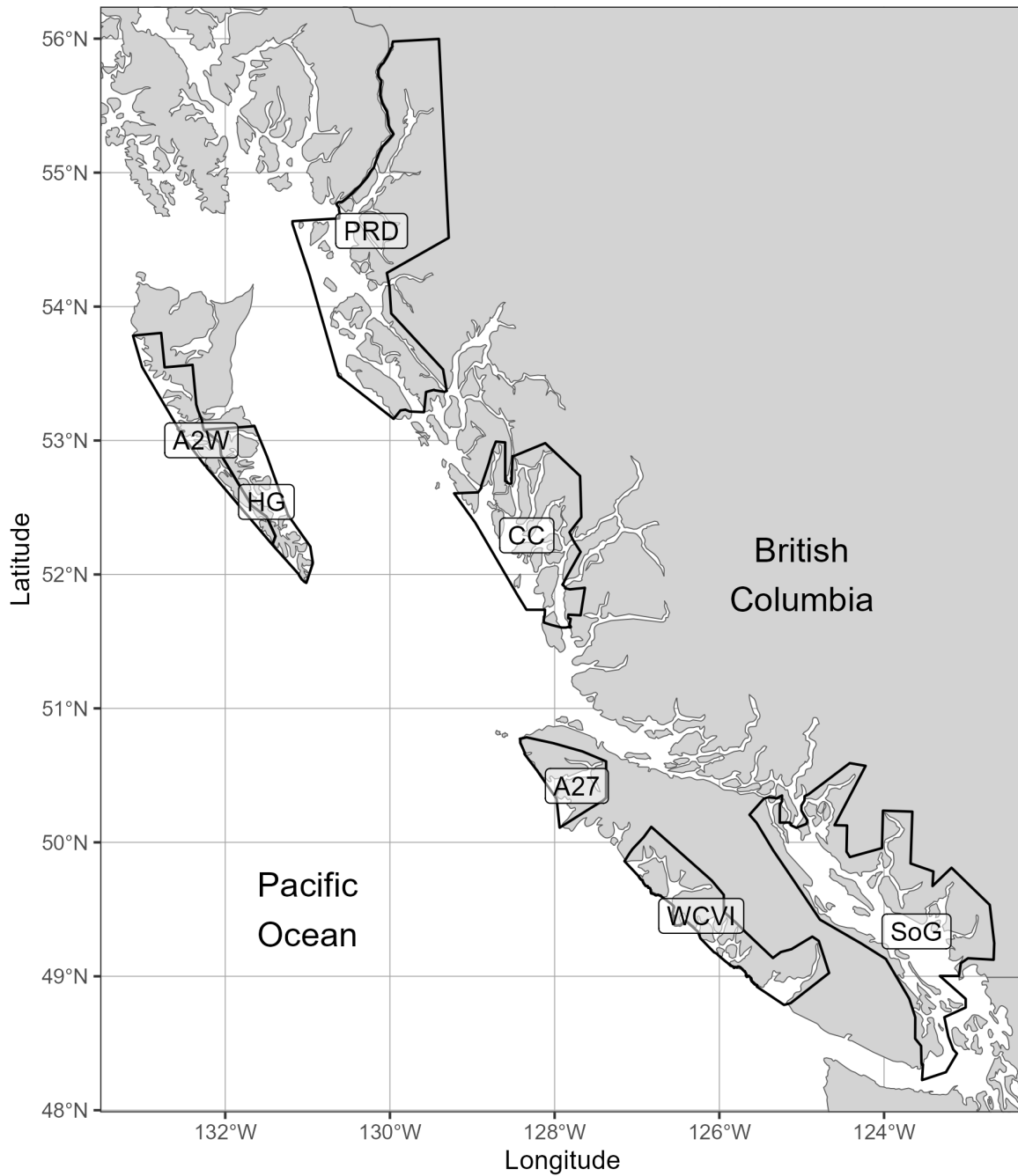


Figure 1. Boundaries for the Pacific Herring stock assessment regions (SARs) in British Columbia. There are 5 major SARs: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SoG), and West Coast of Vancouver Island (WCVI). There are 2 minor SARs: Area 27 (A27) and Area 2 West (A2W). Units: kilometres (km).

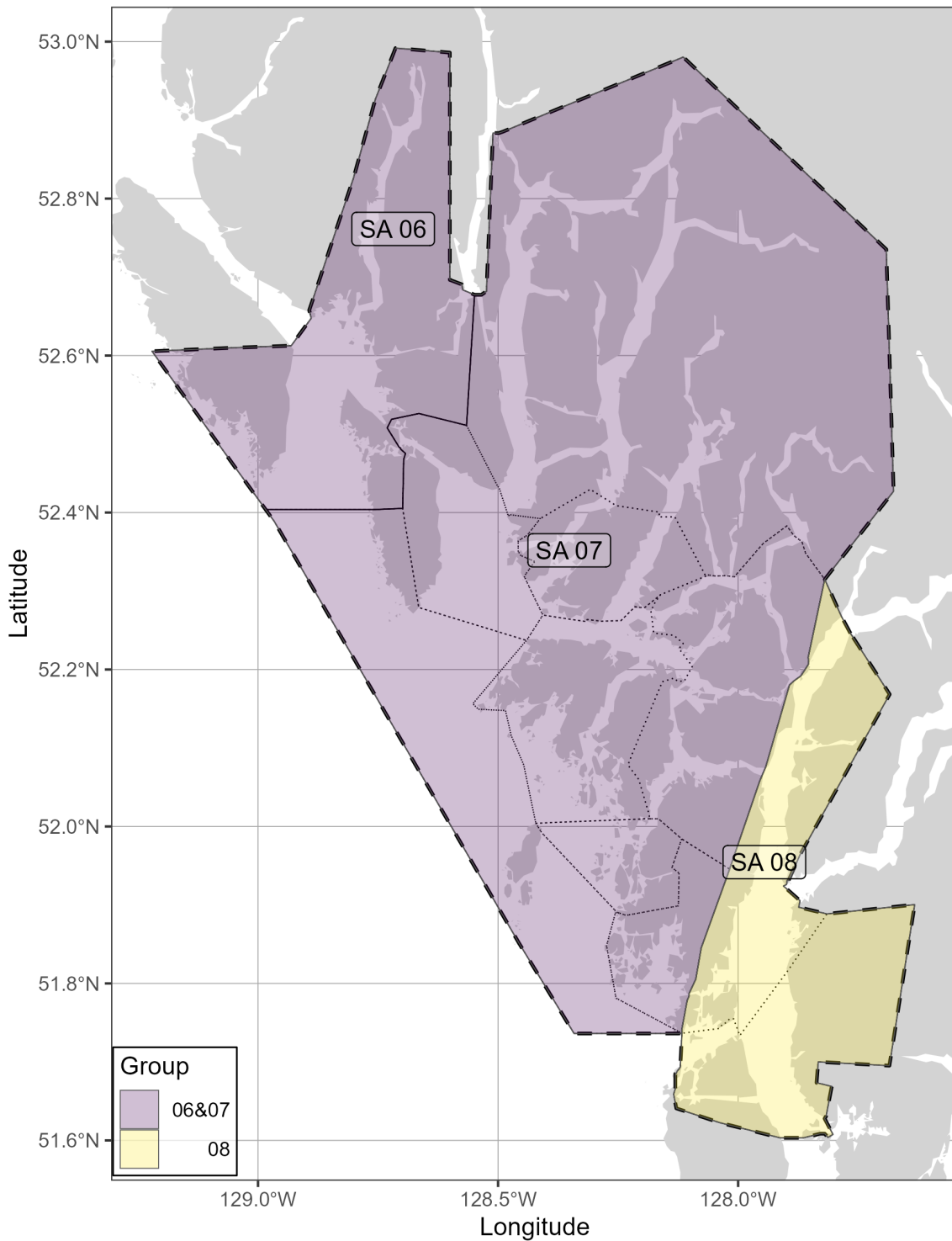


Figure 2. Boundaries for the Central Coast major stock assessment region (SAR; thick dashed lines), associated Statistical Areas (SA; thin solid lines), and associated Sections (thin dotted lines). Units: kilometres (km). Legend: '06&07' is Statistical Areas 06 and 07; and '08' is Statistical Area 08.

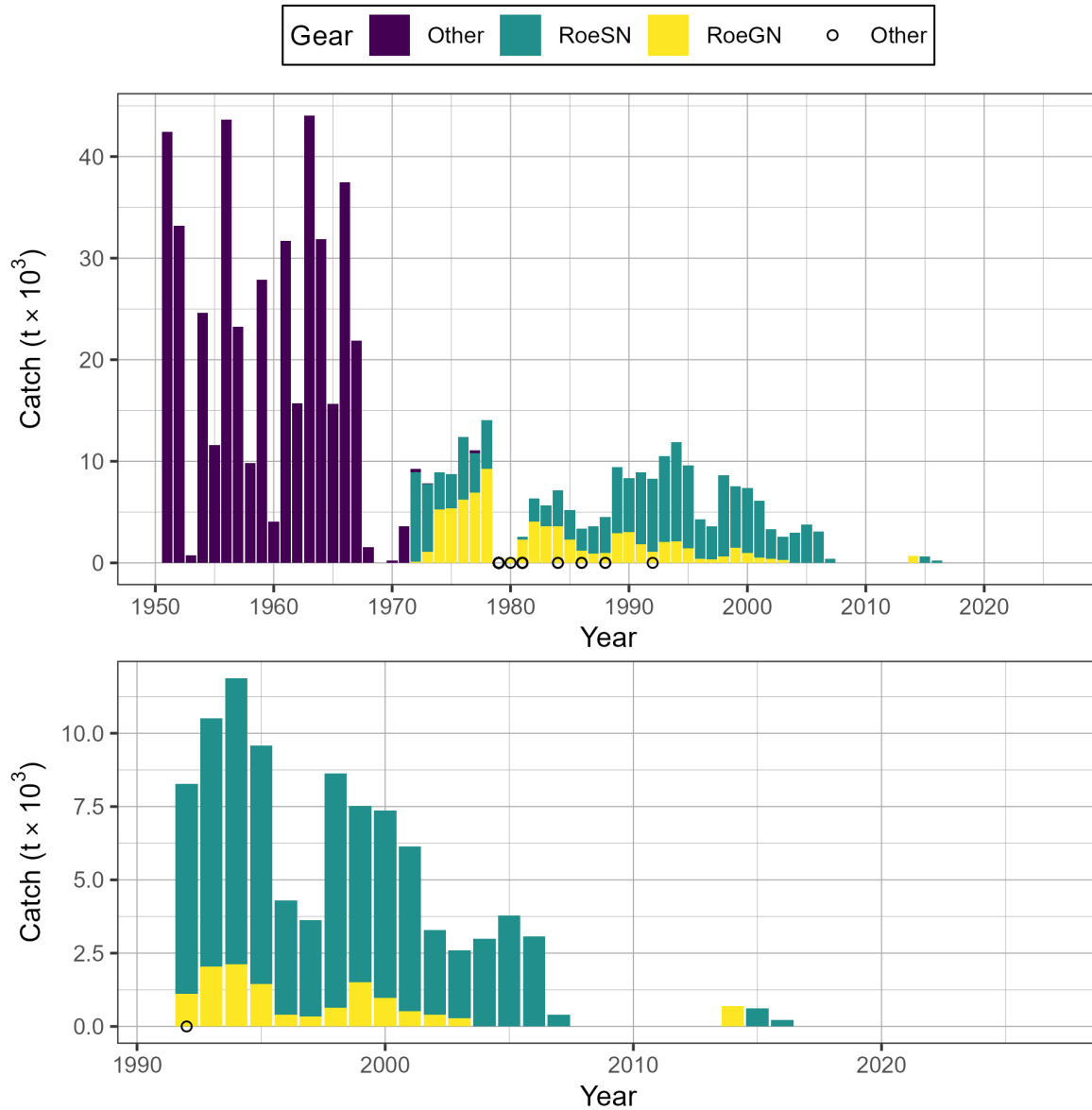


Figure 3. Time series of total landed catch in thousands of metric tonnes ($t \times 10^3$) of Pacific Herring by gear type from 1951 to 2026 in the Central Coast major stock assessment region (SAR). Legend: 'Other' represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; 'RoeSN' represents the roe seine fishery; and 'RoeGN' represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Bottom panel shows catch since 1991 in more detail. Note: symbols indicate years in which catch by gear type (i.e., Other, RoeSN, RoeGN) is withheld due to privacy concerns.

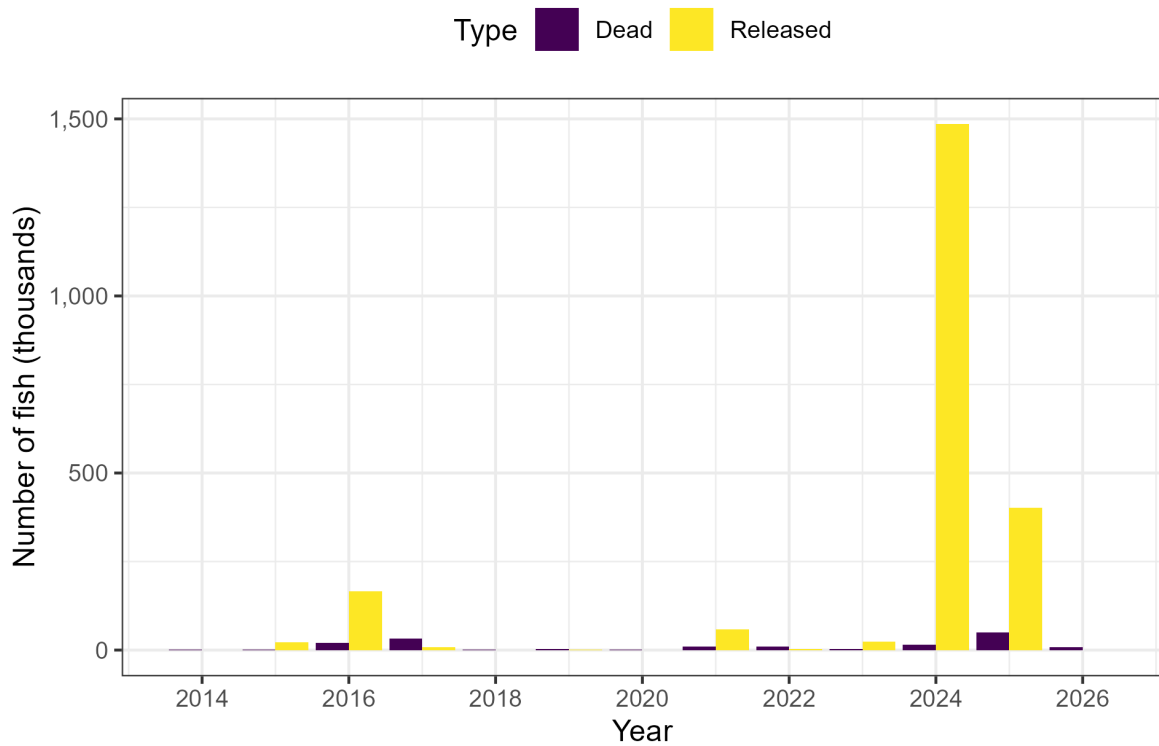


Figure 4. Incidental Pacific Herring mortality in aquaculture activities in thousands of fish from 2014 to 2026 in the Central Coast major stock assessment region (SAR). Note: figure may include data outside SAR boundaries.

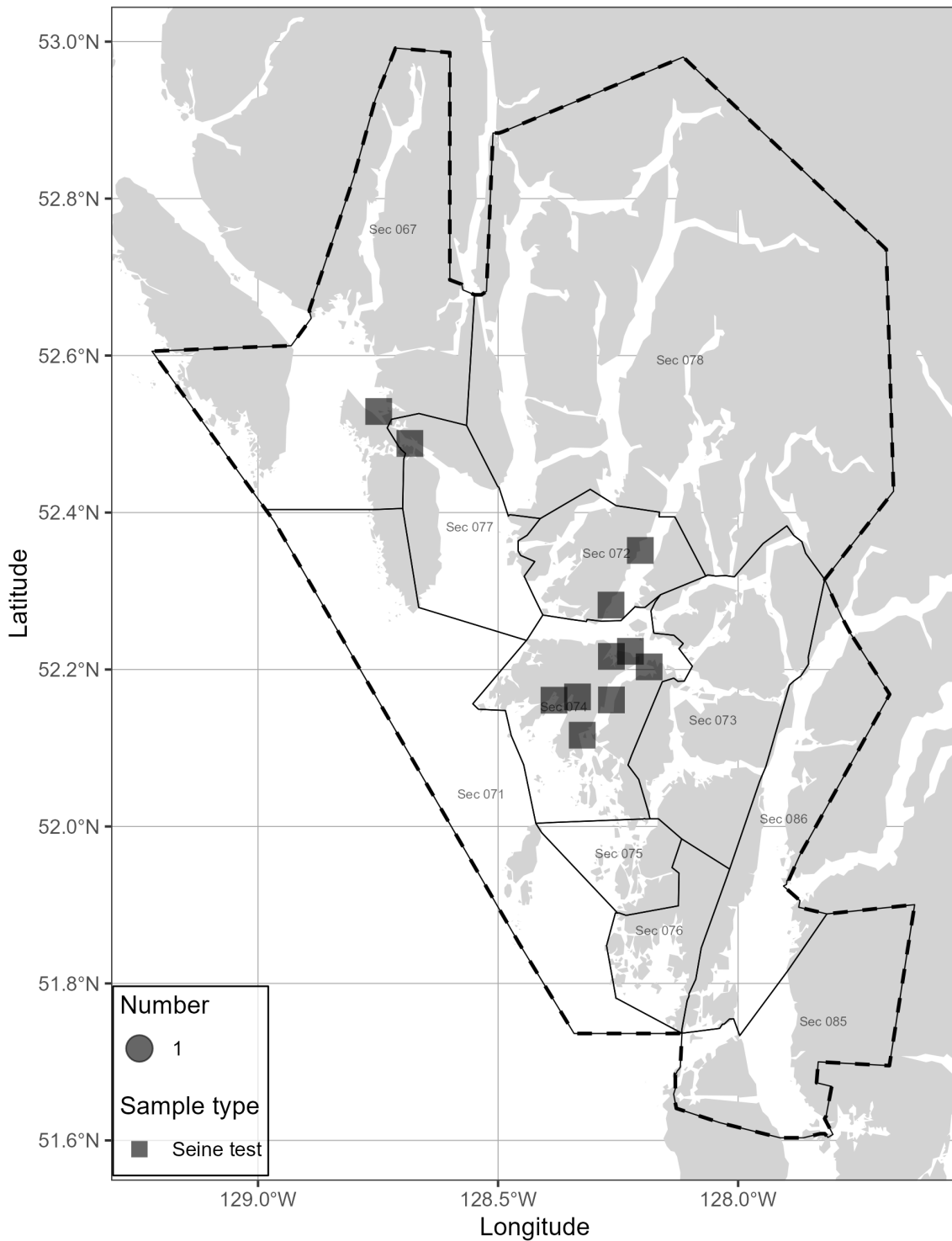


Figure 5. Location and type of Pacific Herring biological samples collected in 2026 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). Units: kilometres (km).

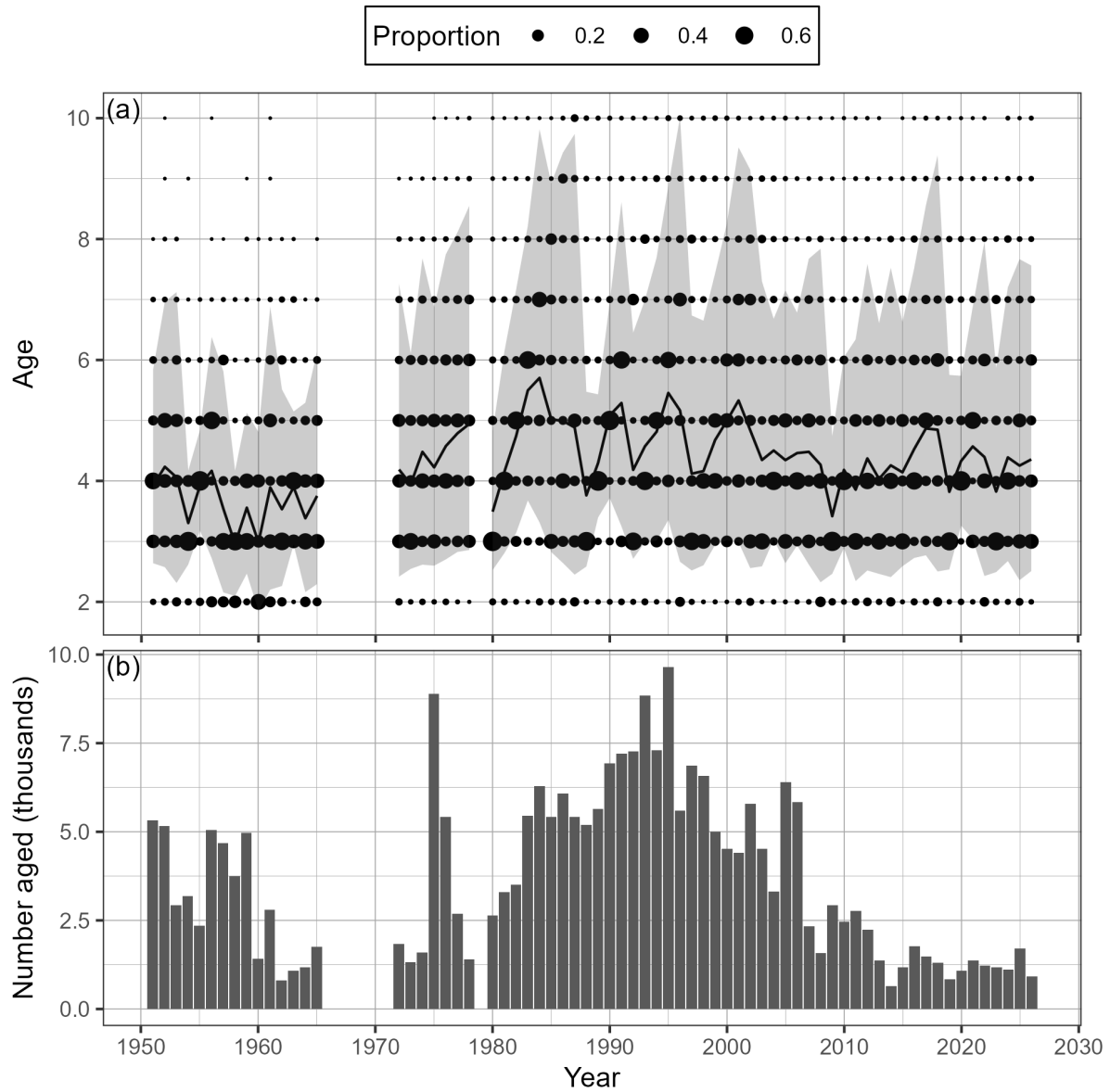


Figure 6. Time series of observed proportion-at-age (a) and number aged in thousands (b) of Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). The black line is the mean age, and the shaded area is the approximate 90% distribution. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

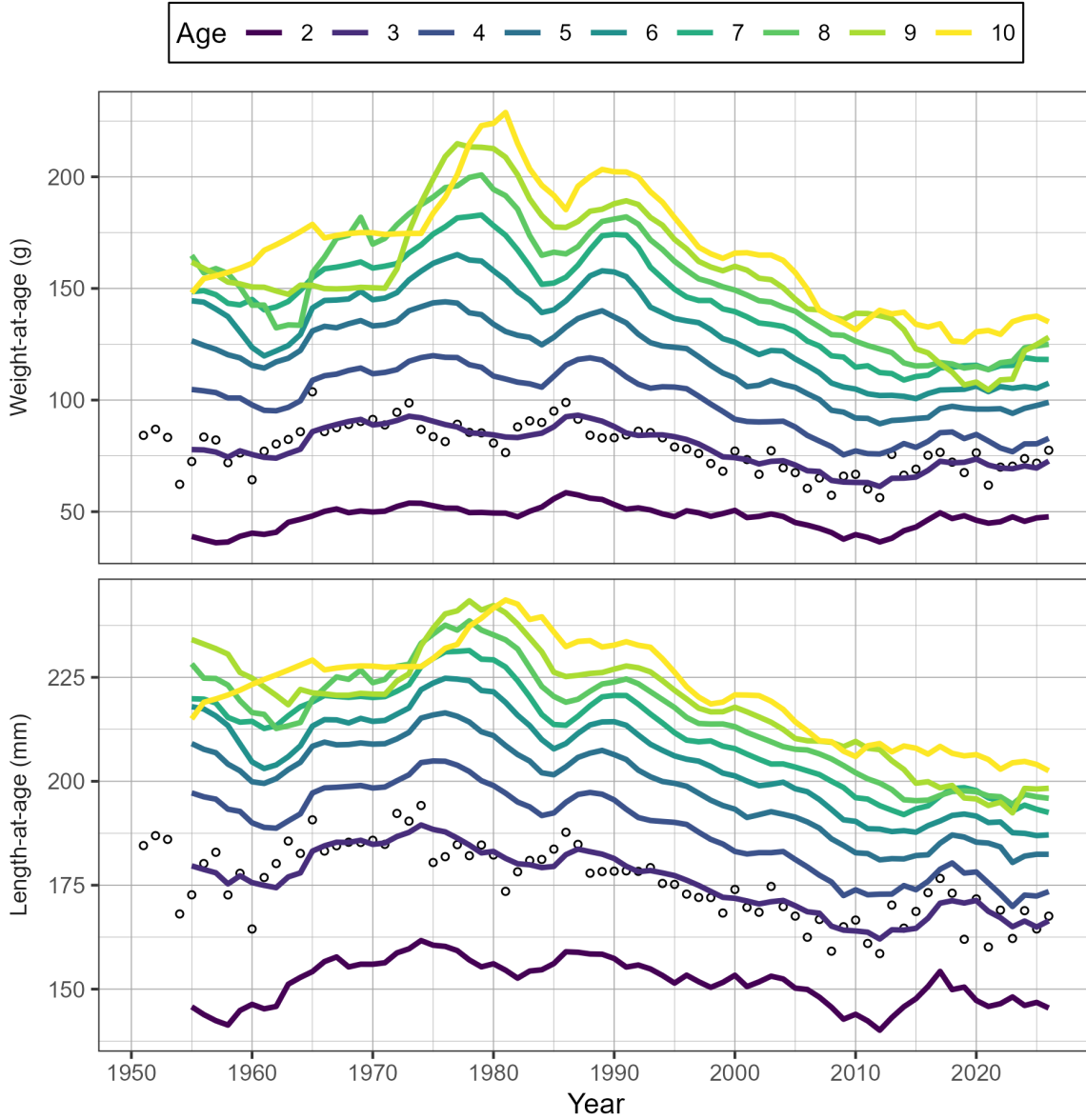


Figure 7. Time series of weight-at-age in grams (g) and length-at-age in millimetres (mm) for age-3 (circles) and 5-year running mean weight- and length-at-age (lines) for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). Missing weight- and length-at-age values (i.e., years with no biological samples) are imputed using one of two methods: missing values at the beginning of the time series are imputed by extending the first non-missing value backwards; other missing values are imputed as the mean of the previous 5 years. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

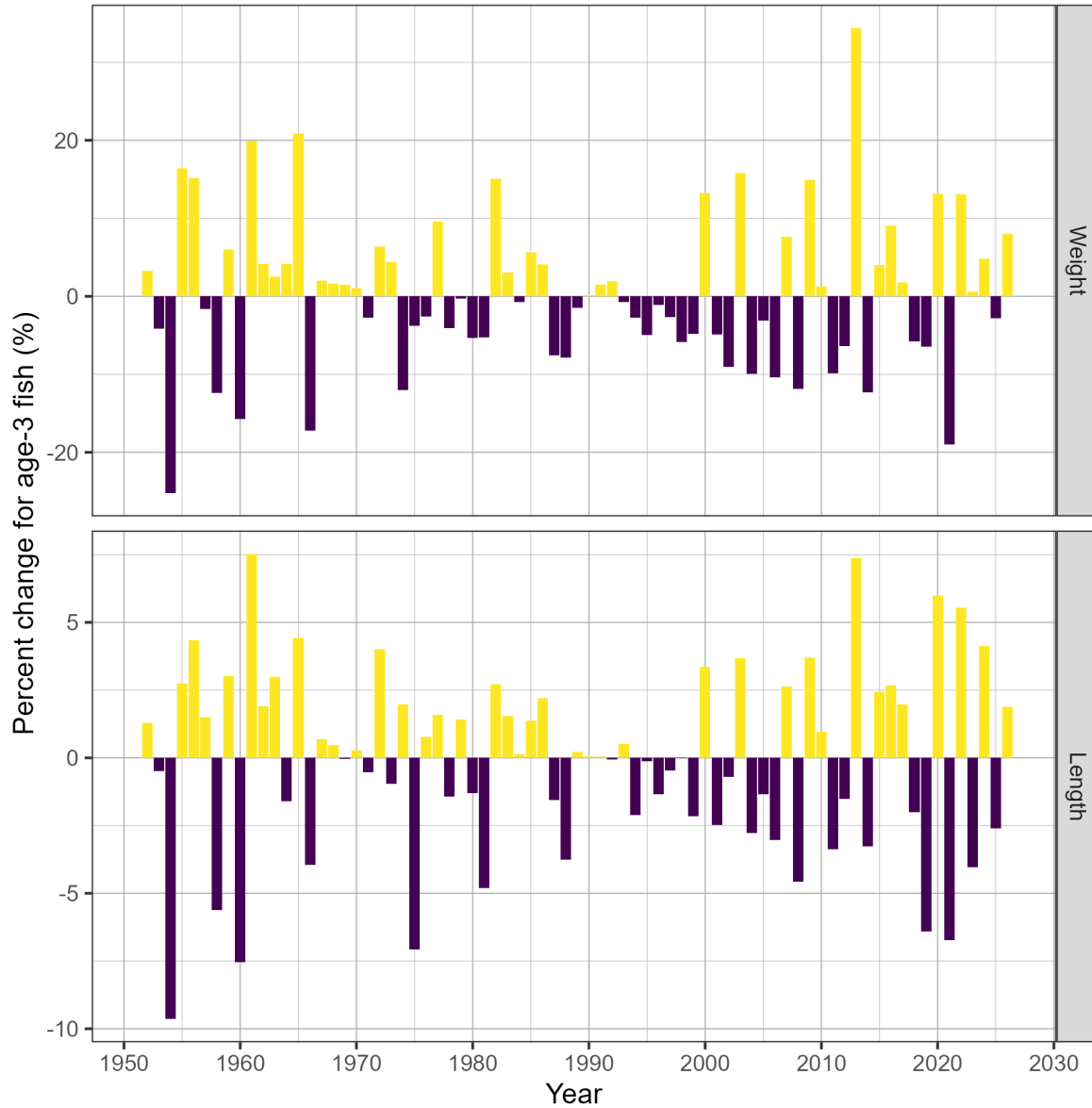


Figure 8. Time series of percent change (%) in weight and length for age-3 fish for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-3 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

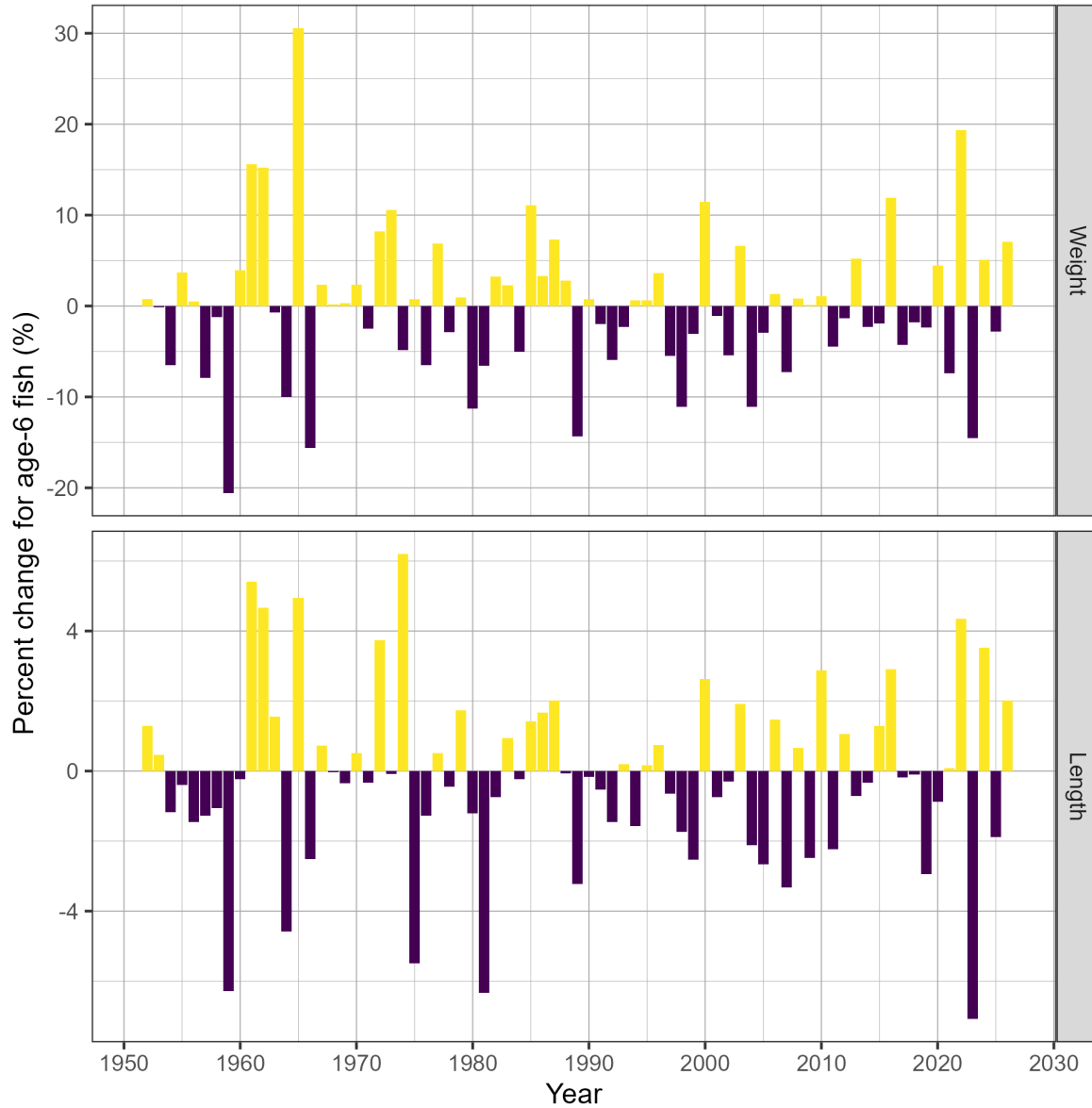


Figure 9. Time series of percent change (%) in weight and length for age-6 fish for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-6 fish, respectively, in year t . Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

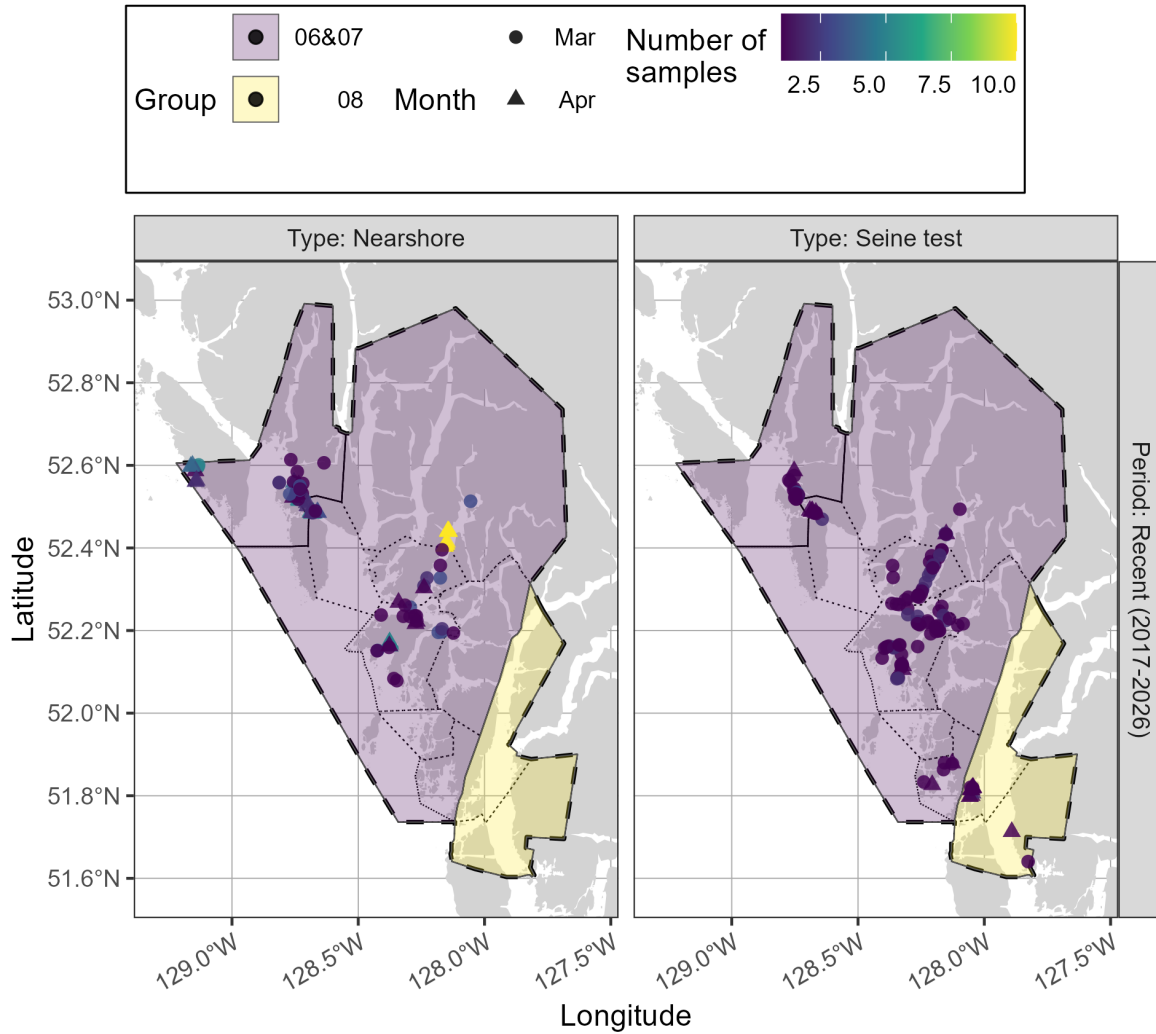


Figure 10. Location of Pacific Herring biological samples in the Central Coast major stock assessment region (SAR) by period, sample type, and month. Legend: ‘Nearshore’ refers to samples collected using cast nets as part of a pilot study with First Nations.

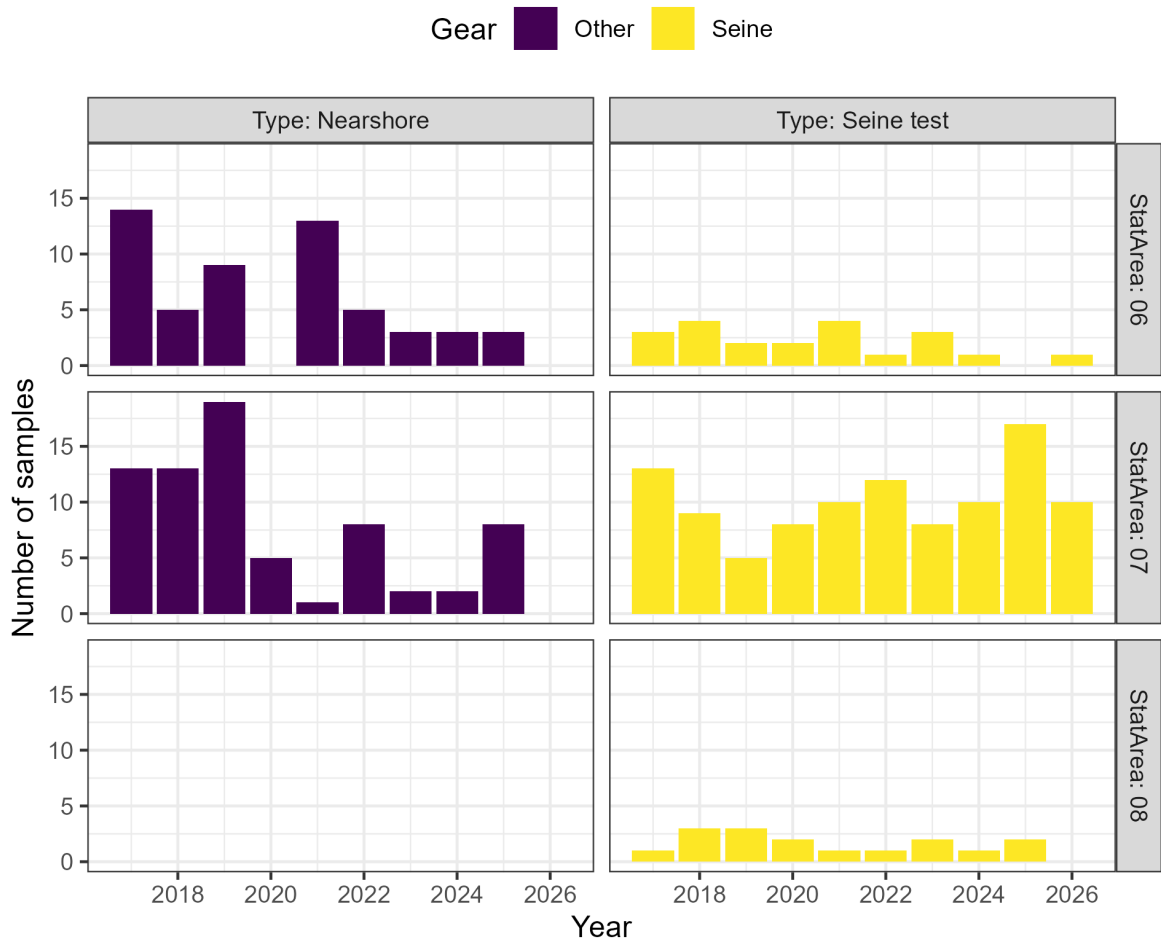


Figure 11. Number of Pacific Herring samples in the Central Coast major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

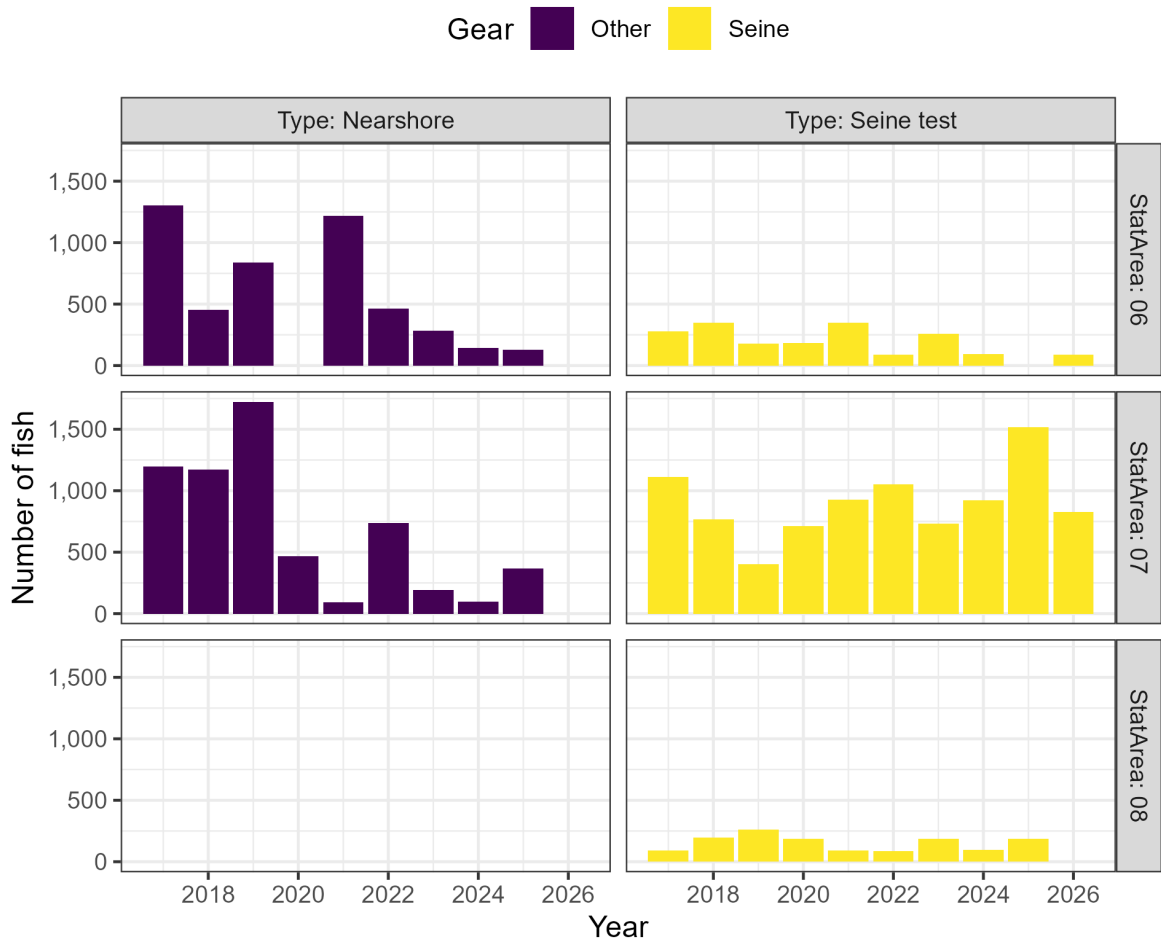


Figure 12. Number of Pacific Herring in biological samples in the Central Coast major stock assessment region (SAR) by sample type, statistical area, and gear type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

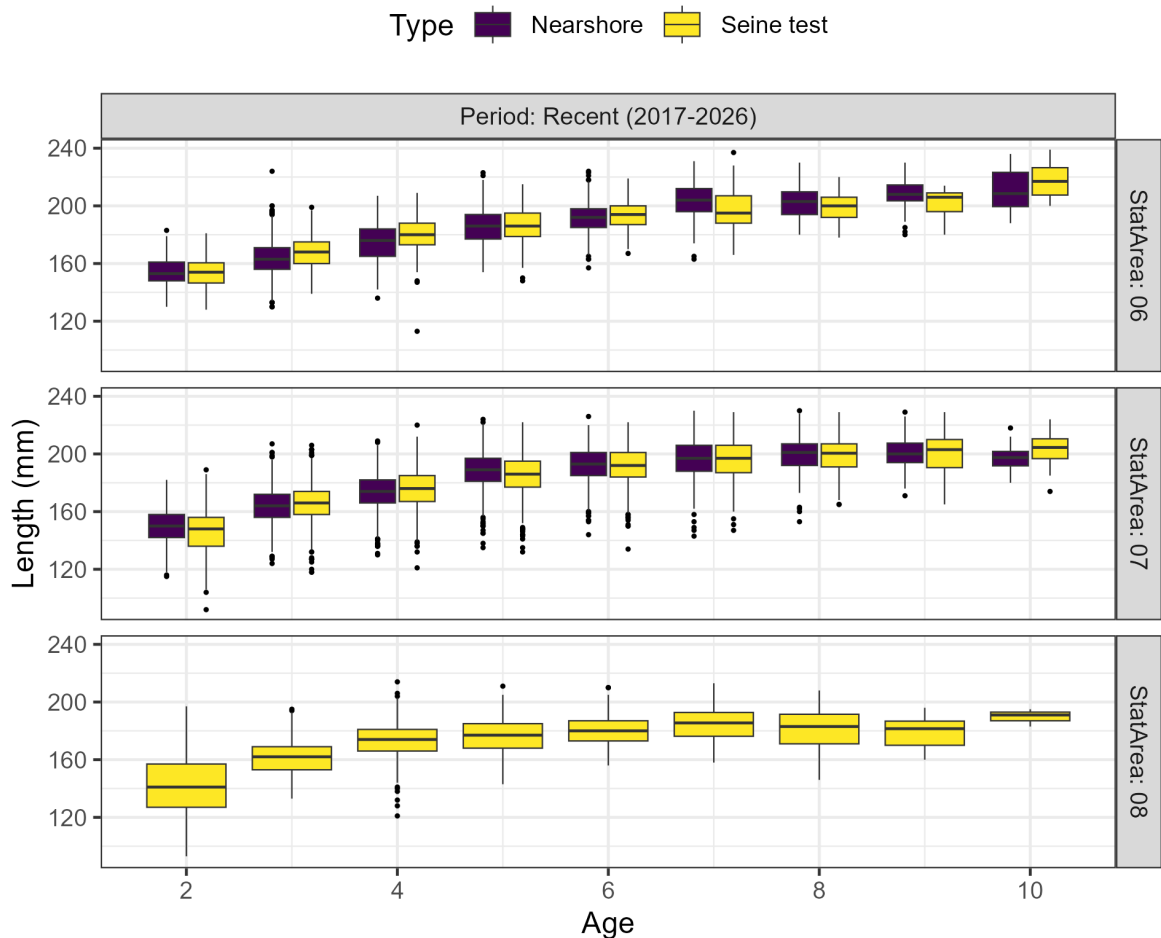


Figure 13. Length-at-age in millimetres (mm) of Pacific Herring in the Central Coast major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times \text{IQR}$, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

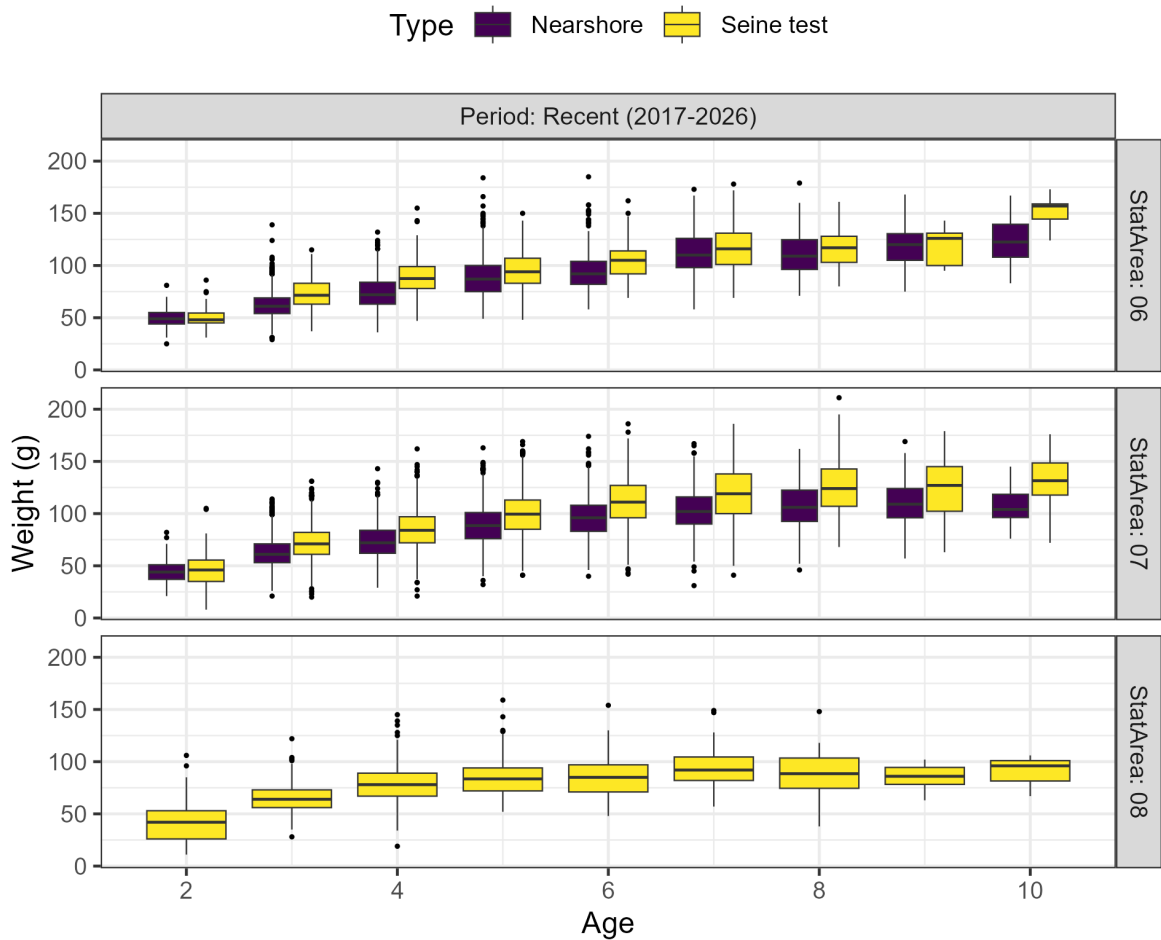


Figure 14. Weight-at-age in grams (g) of Pacific Herring in the Central Coast major stock assessment region (SAR) by period, statistical area, and sample type. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times \text{IQR}$, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

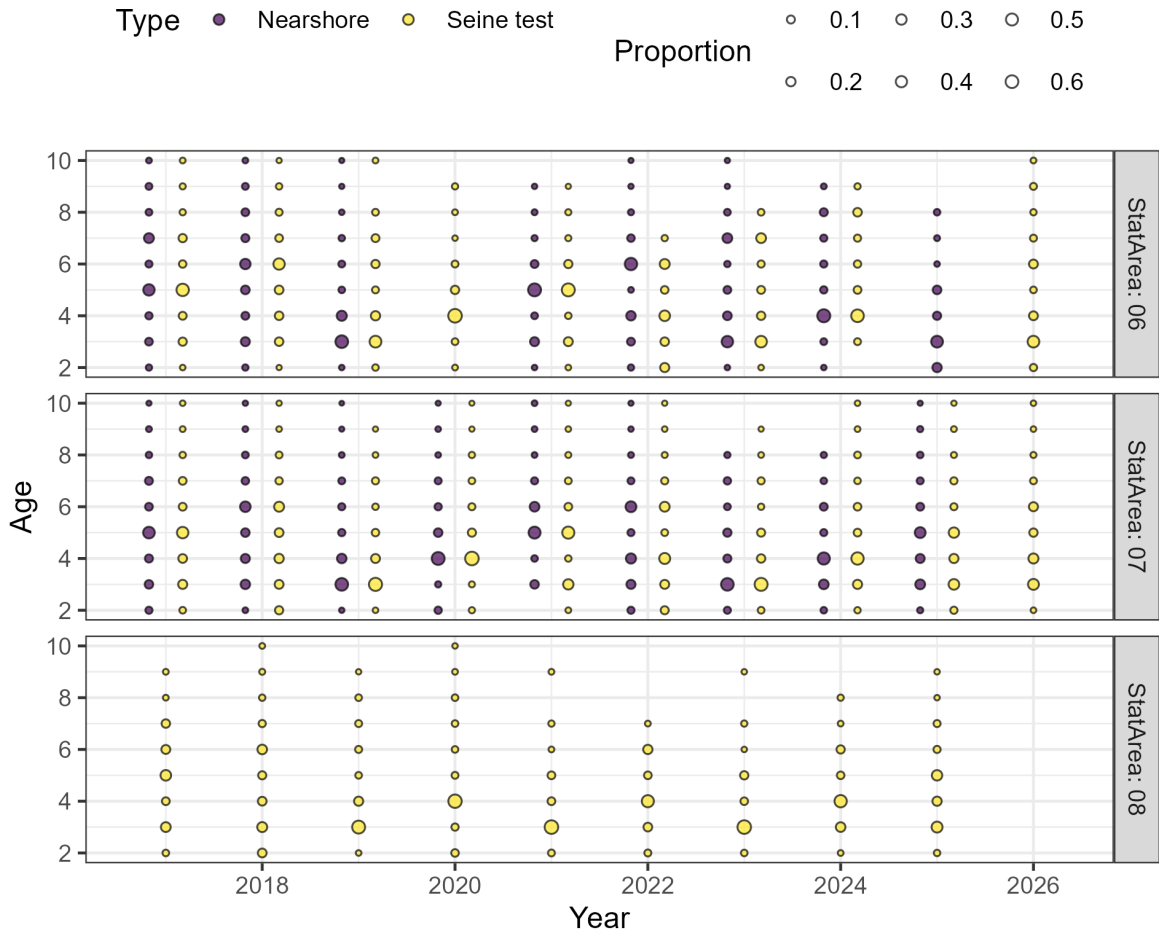


Figure 15. Proportion-at-age of Pacific Herring in the Central Coast major stock assessment region (SAR) by statistical area and sample type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations. The age-10 class is a 'plus group' which includes fish ages 10 and older.

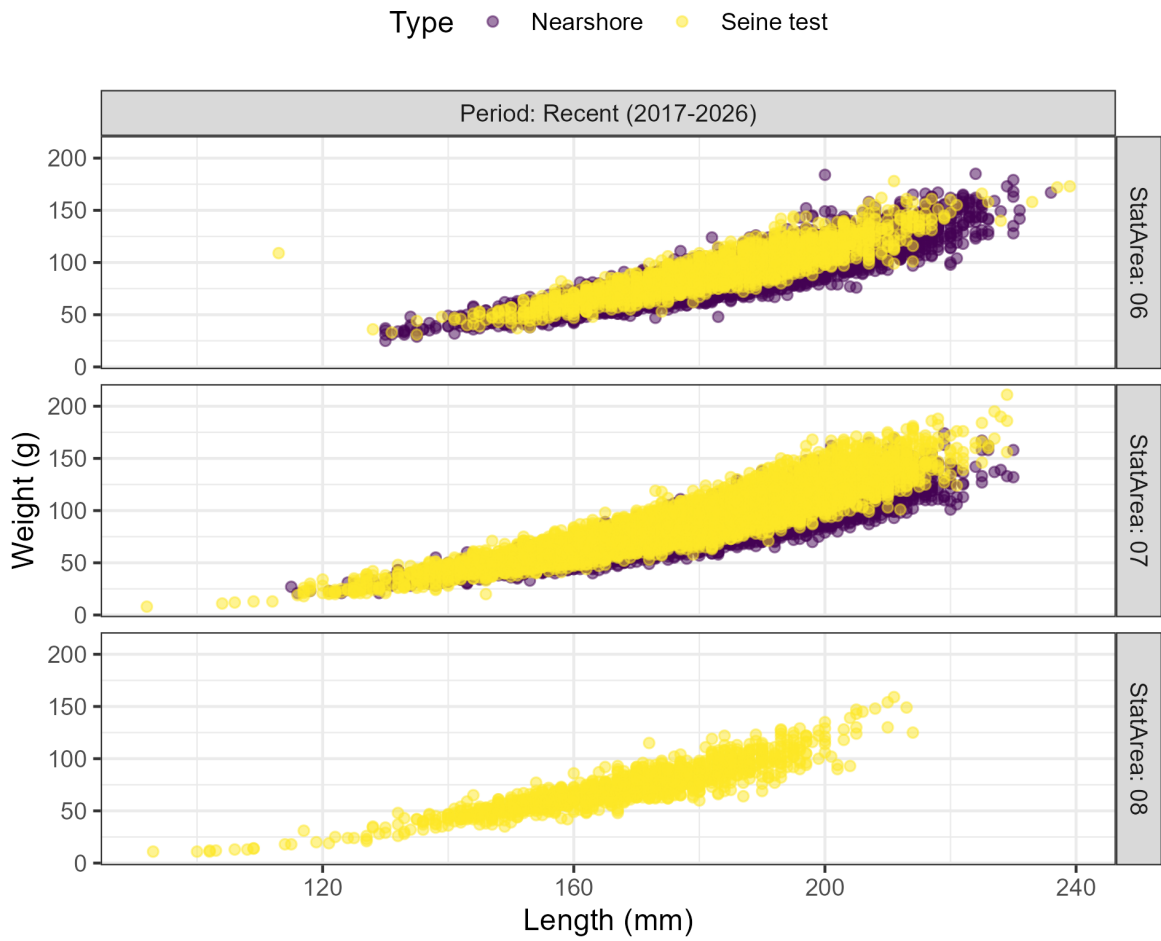


Figure 16. Length-weight relationship for Pacific Herring in the Central Coast major stock assessment region (SAR) by period, statistical area, and sample type. Legend: 'Nearshore' refers to samples collected using cast nets as part of a pilot study with First Nations.

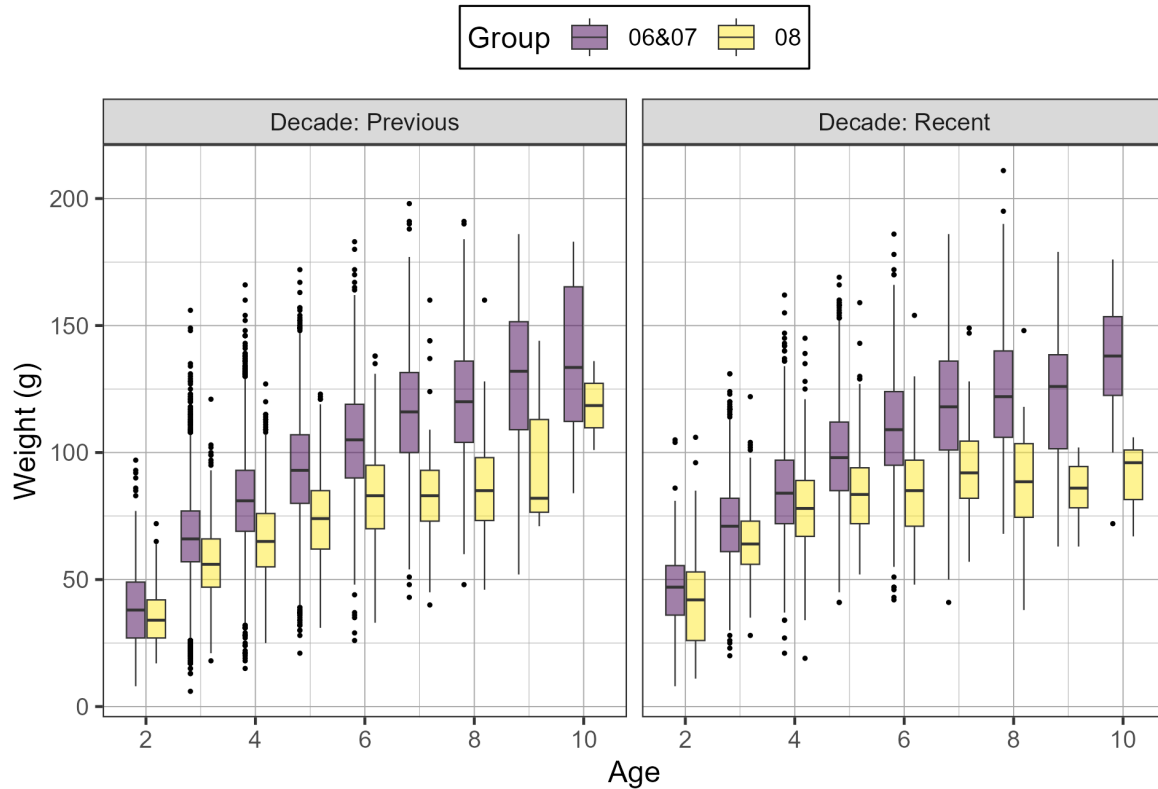


Figure 17. Weight-at-age in grams (g) of Pacific Herring in the Central Coast major stock assessment region (SAR) by Group from the most recent decade (2017 to 2026), and the previous decade (2007 to 2016). The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to $1.5 \times \text{IQR}$, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Sample sizes are given in Table 16. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older. Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

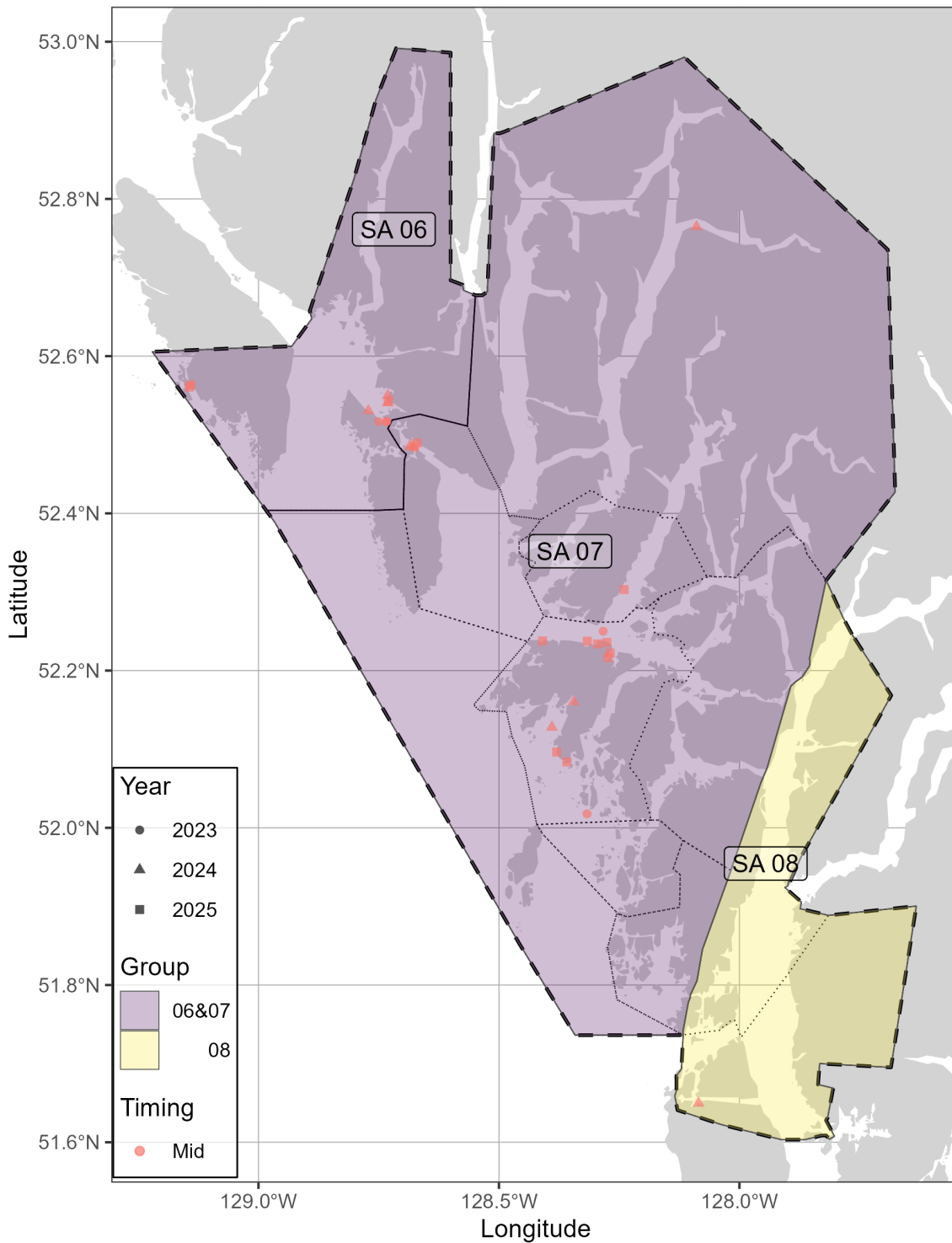


Figure 18. Nearshore genetics sampling in the Central Coast major SAR. Note that timing indicates the collection date: ‘Early’ is January and February, ‘Mid’ is March and April, and ‘Late’ is May, June, and July. In addition, note that genetic samples from 2025 are not yet available. See Figure 2 for legend.

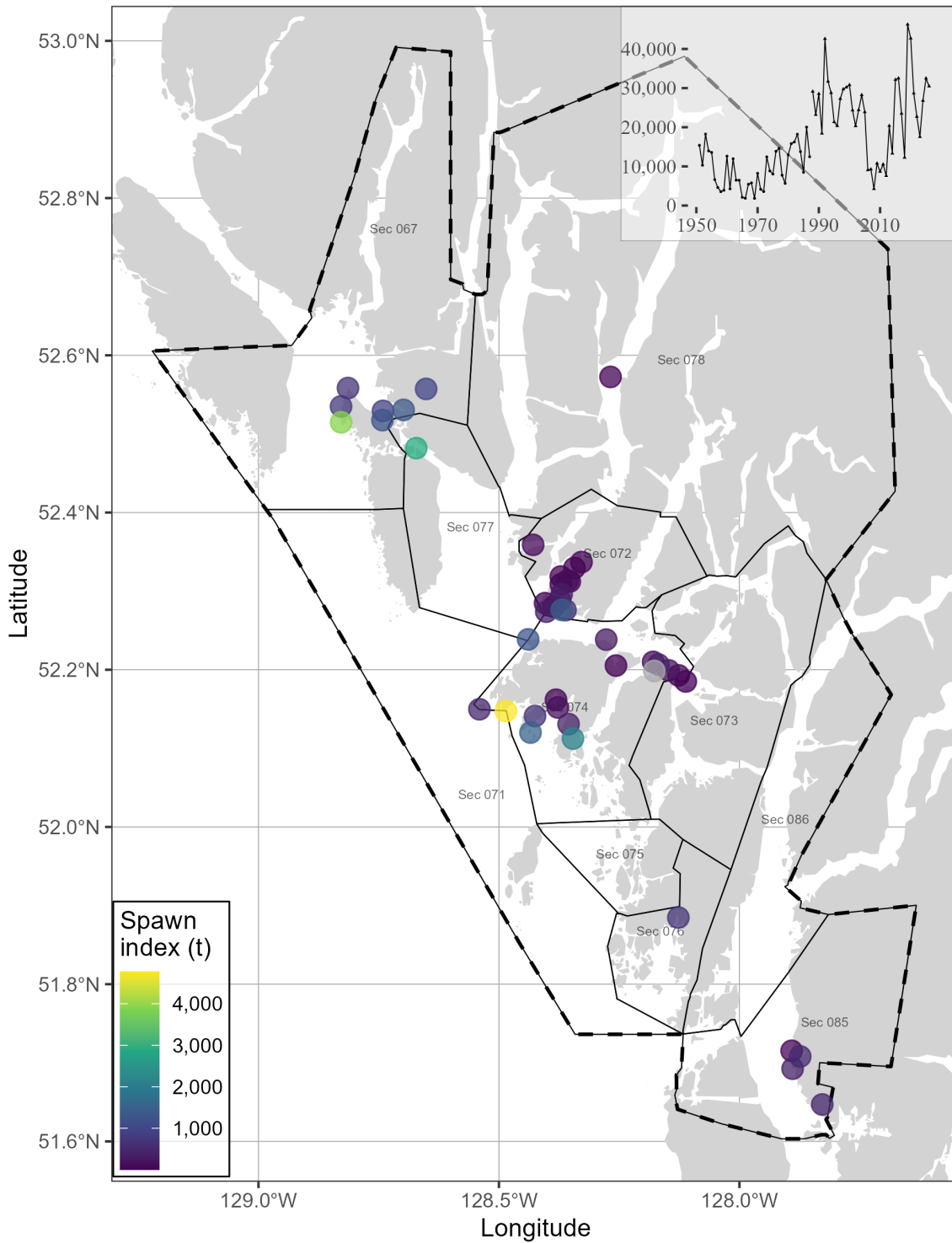


Figure 19. Pacific Herring spawn survey locations, and spawn index in metric tonnes (t) in 2026 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km).

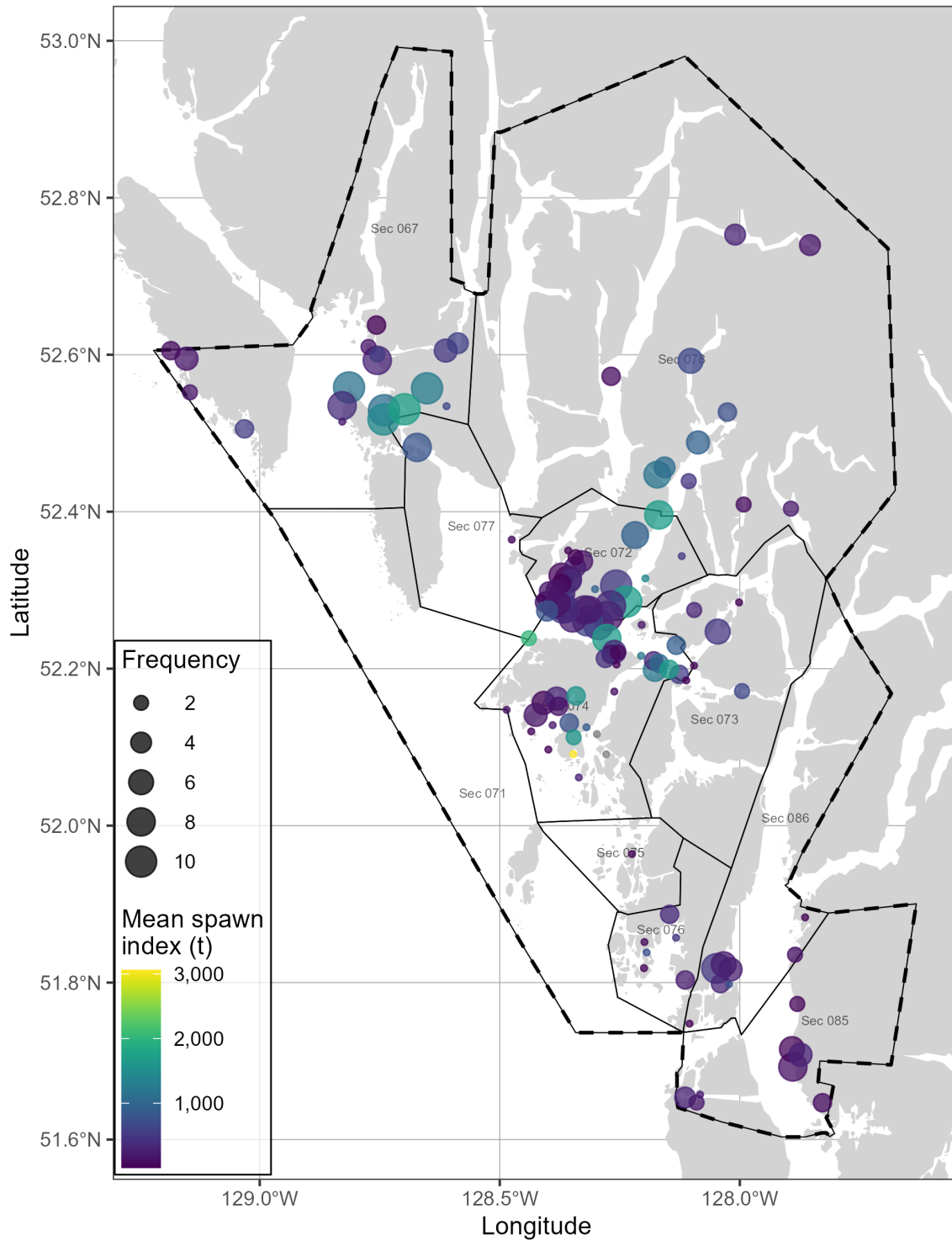


Figure 20. Pacific Herring spawn survey locations, mean spawn index in metric tonnes (t), and spawn frequency from 2016 to 2025 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Units: kilometres (km).

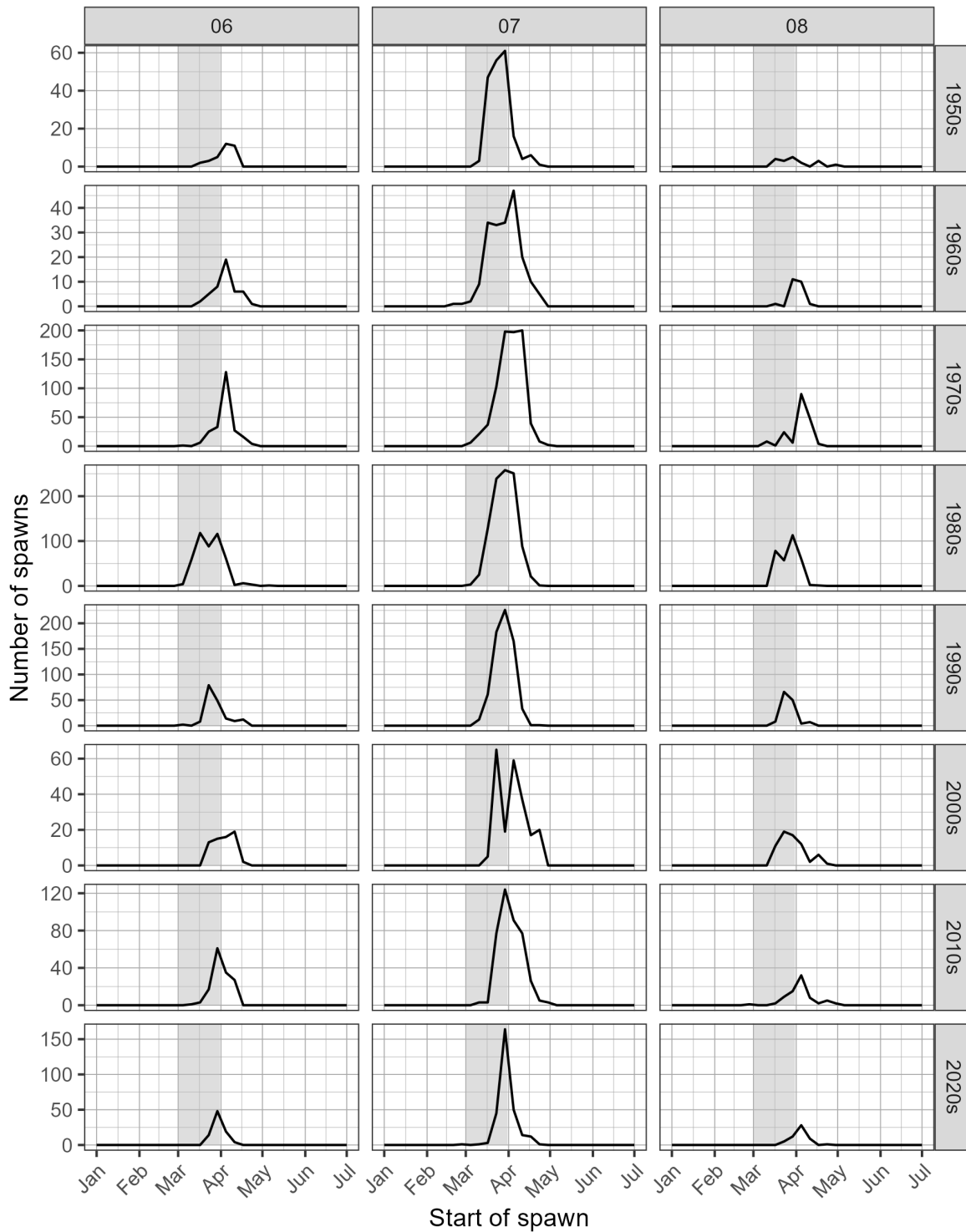


Figure 21. Pacific Herring spawn start date by decade and Statistical Area. Grey shaded regions indicate March 1st to 31st. Note that spawn size and intensity varies; therefore the number of spawns is not directly proportional to spawn extent or biomass.

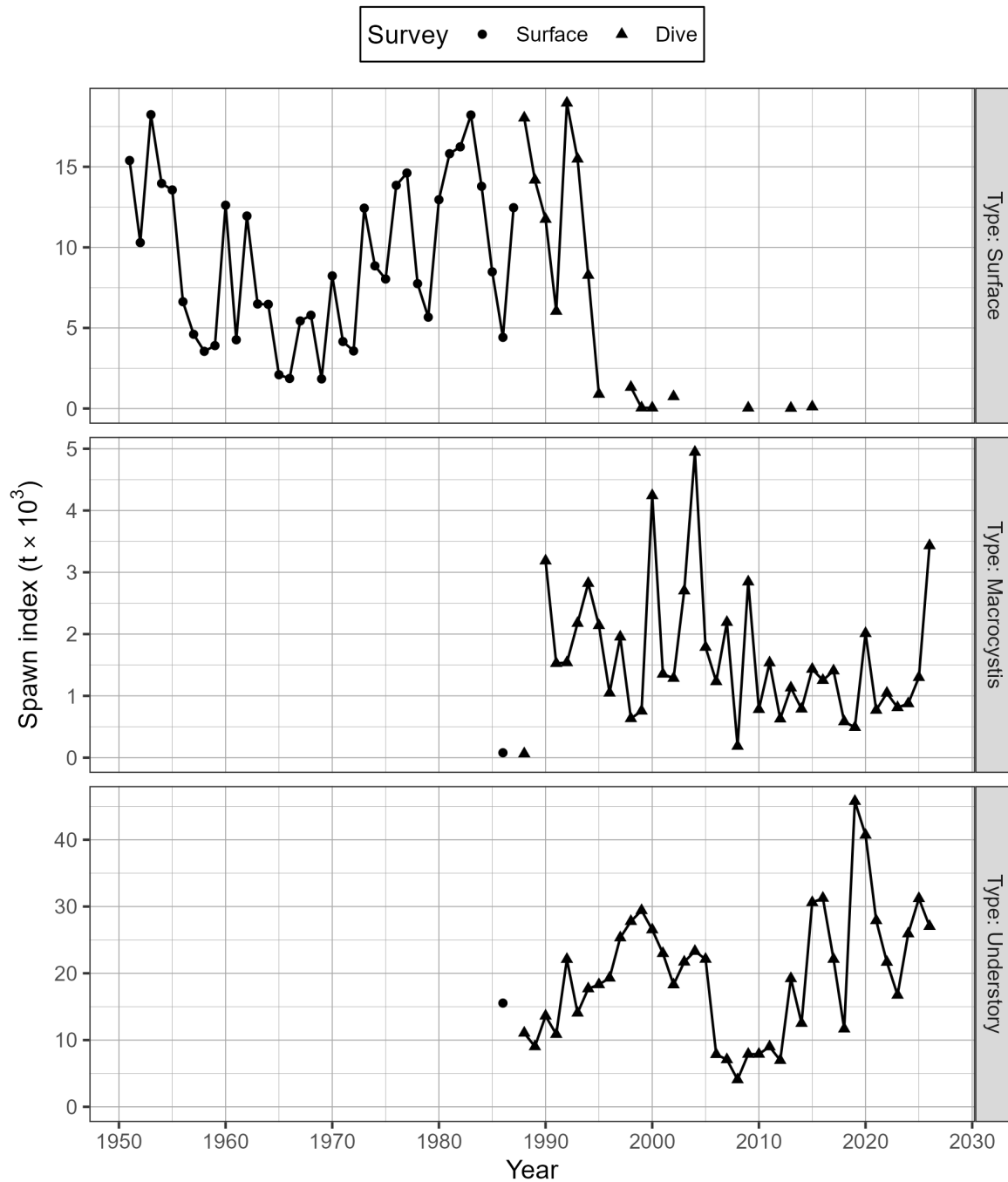


Figure 22. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) by type for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). There are three types of spawn survey observations: observations of spawn taken from the surface usually at low tide, underwater observations of spawn on giant kelp, *Macrocystis* (*Macrocystis* spp.), and underwater observations of spawn on other types of algae and the substrate, which we refer to as ‘understory.’ The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). See Figure 23 for the spawn survey method.

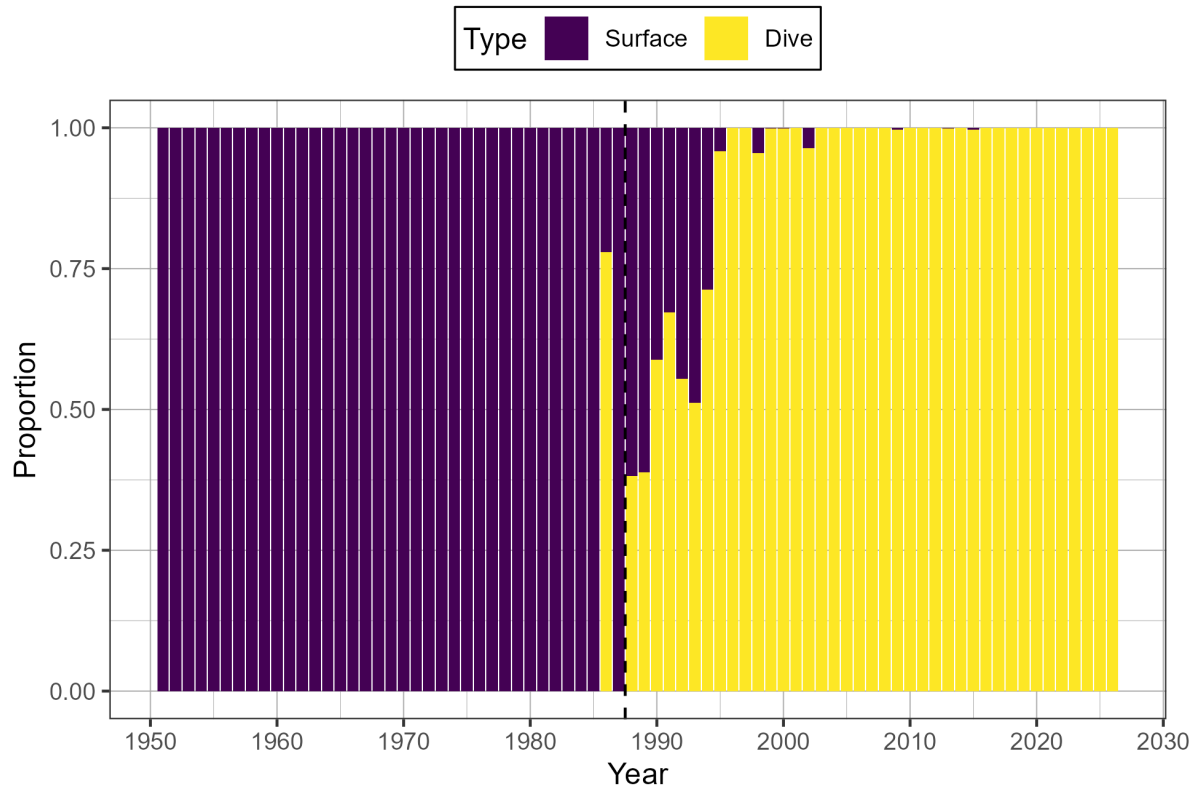


Figure 23. Time series of proportion of spawn index by surface and dive surveys for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

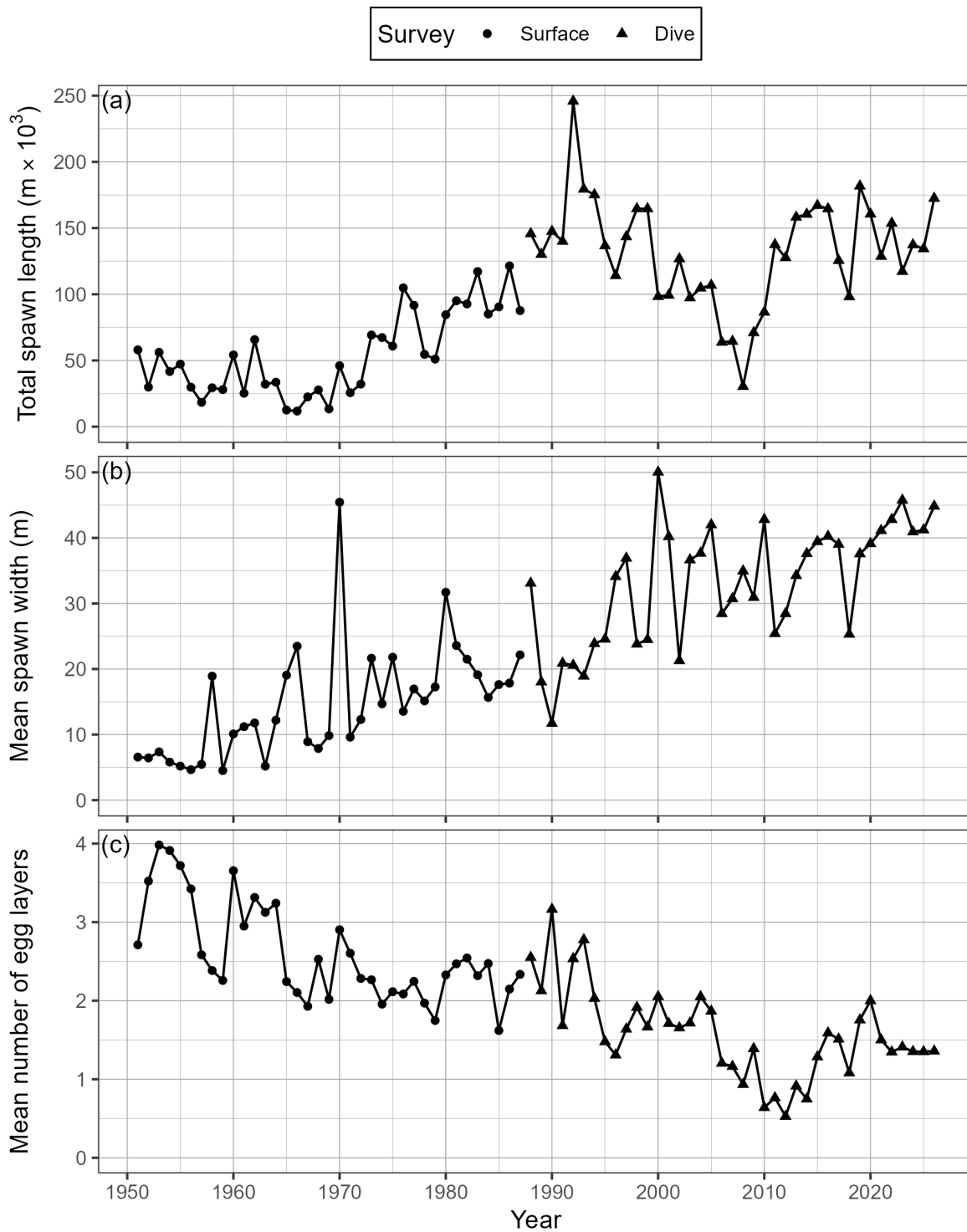


Figure 24. Time series of total spawn length in thousands of metres ($m \times 10^3$; panel a), mean spawn width in metres (b), and mean number of egg layers (c) for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

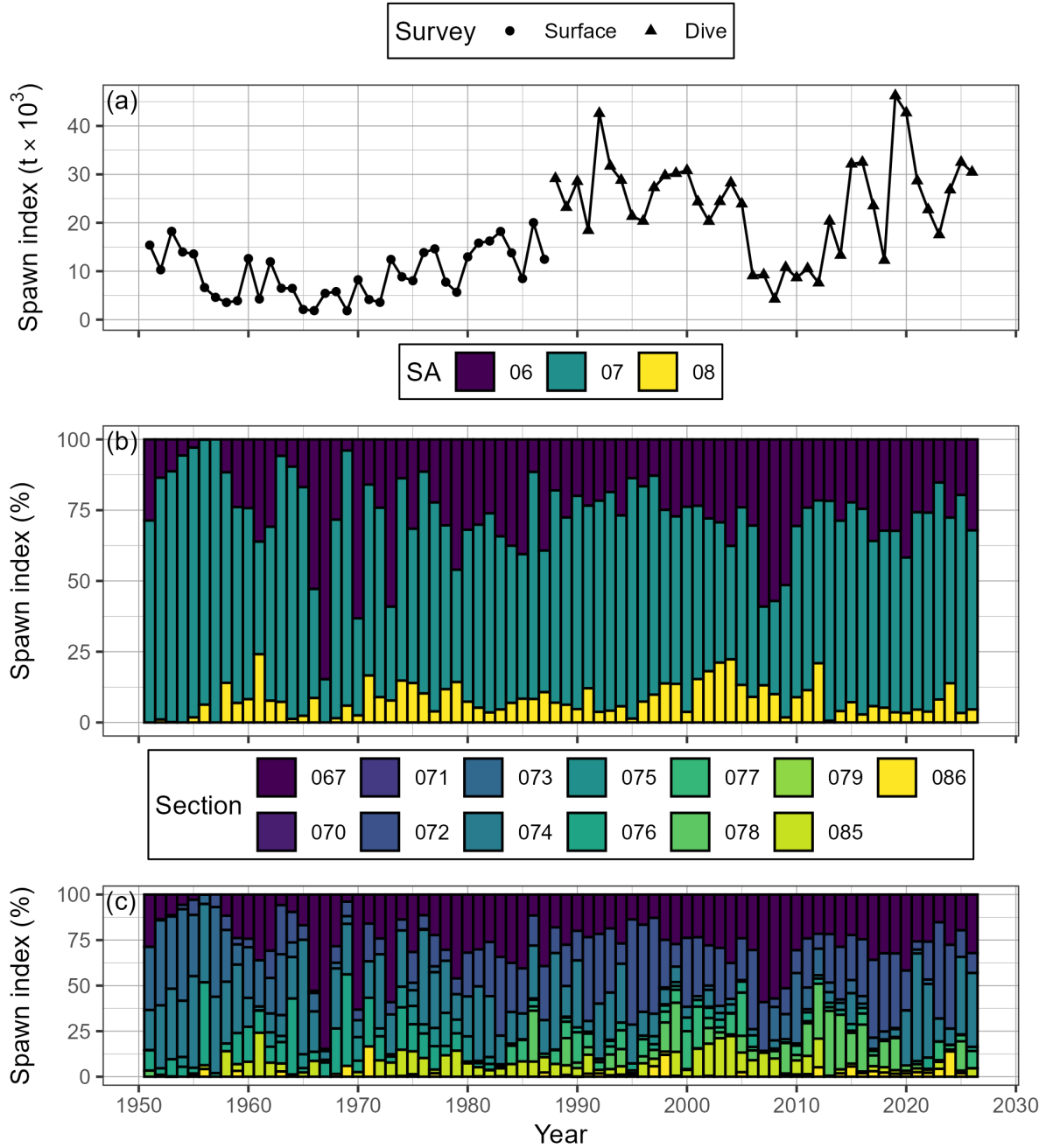


Figure 25. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR; panel a), as well as percent contributed by Statistical Area (SA), and Section (b, & c, respectively). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 22 and 23). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

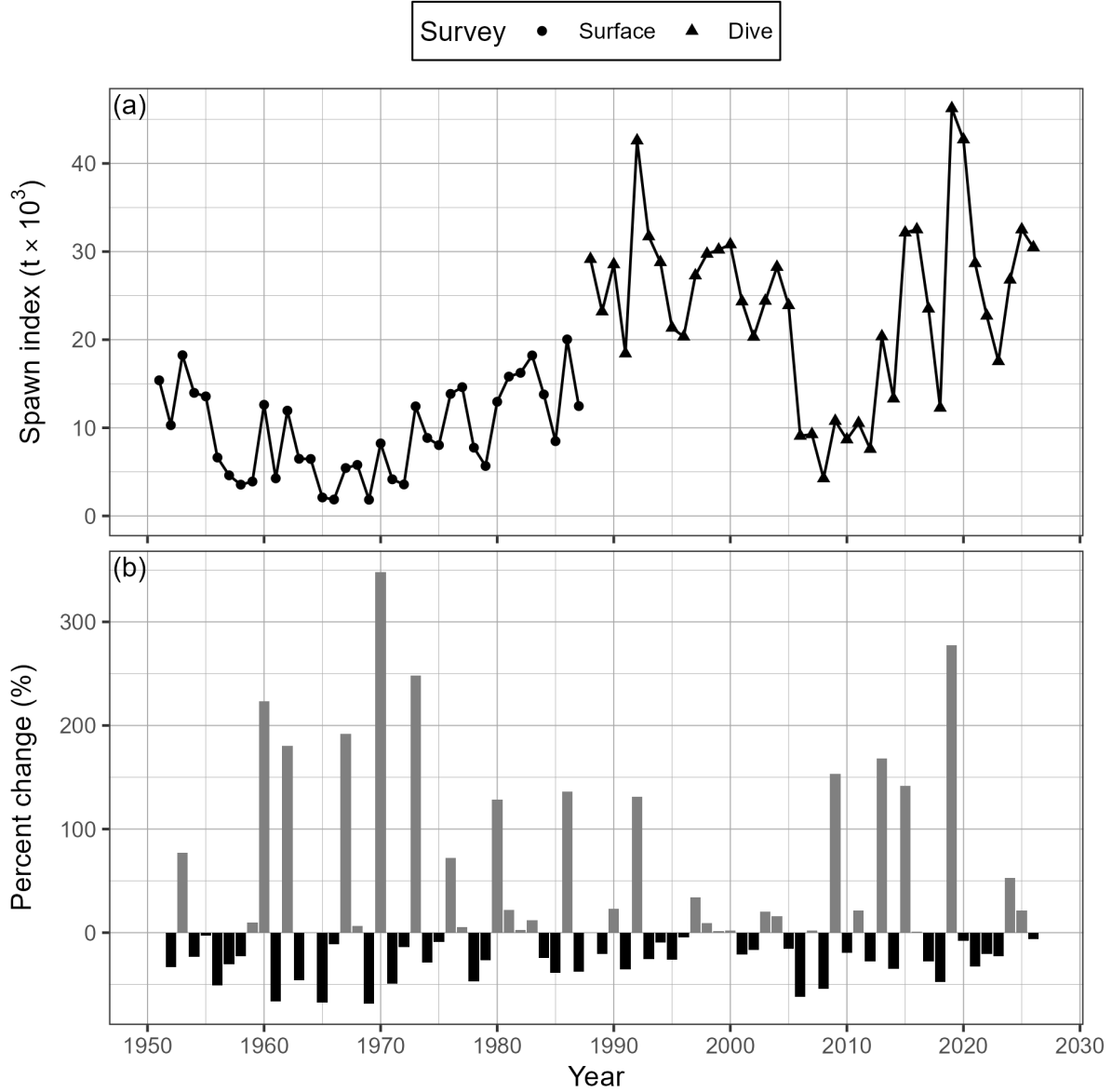


Figure 26. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR; panel a), and percent change (b). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the spawn index in year t . The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 22 and 23).

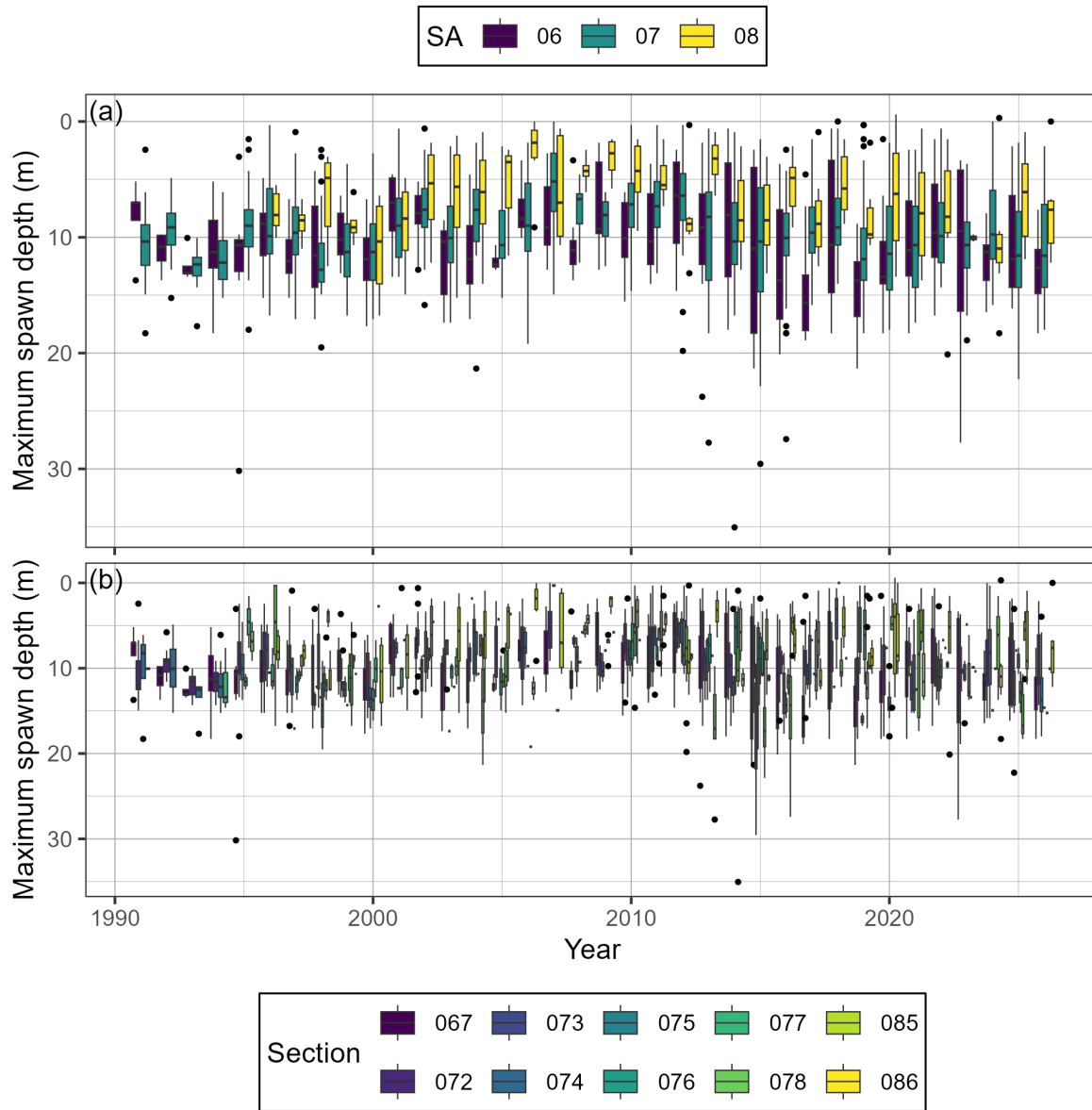


Figure 27. Time series of maximum spawn depth in metres (m) for Pacific Herring from 1991 to 2026 in the Central Coast major stock assessment region (SAR) by Statistical Area (SA; panel a), and Section (b). Note that depth is not corrected to chart datum. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

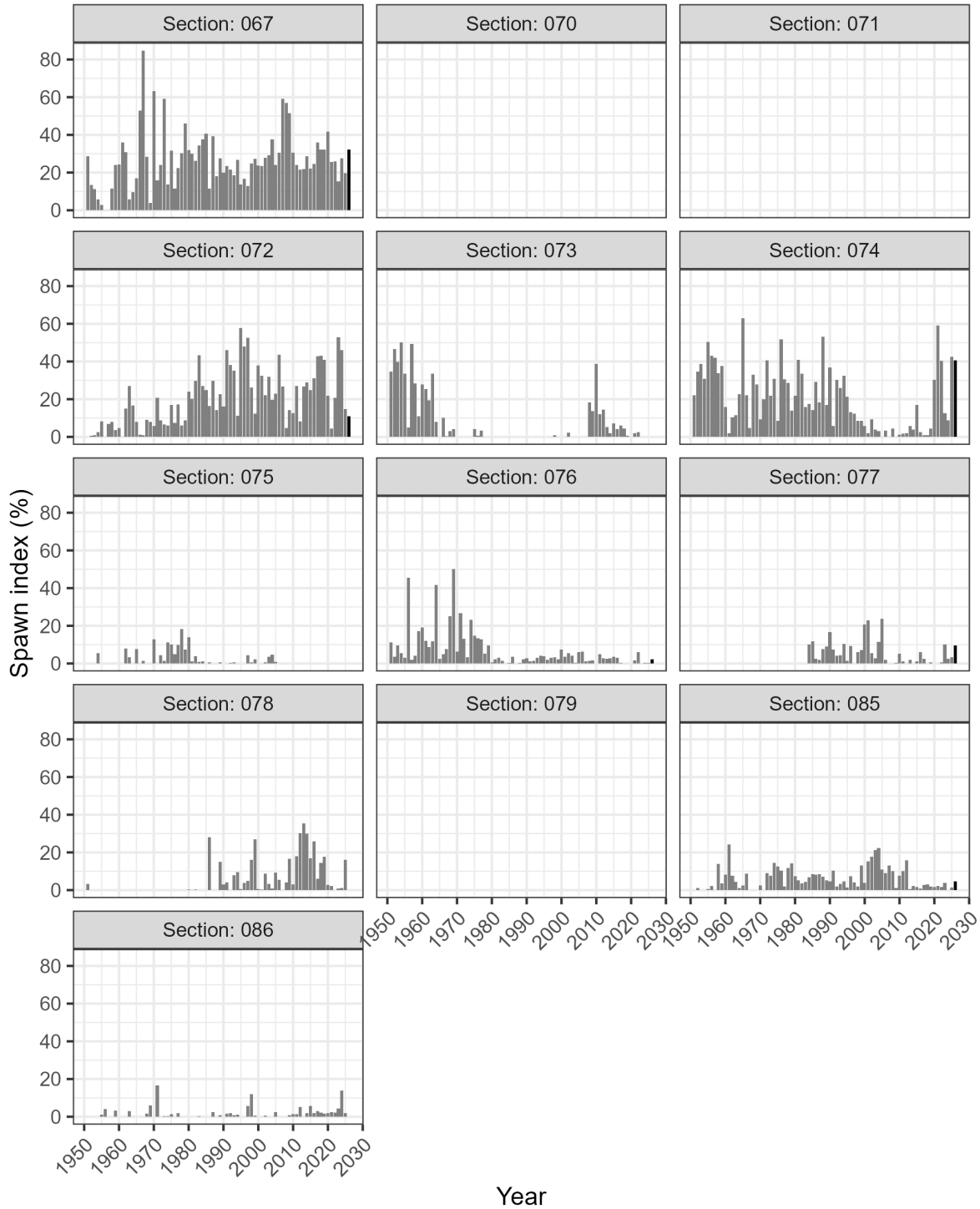


Figure 28. Time series of percent of spawn index by Section for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). The year 2026 has a darker bar to facilitate interpretation. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q .

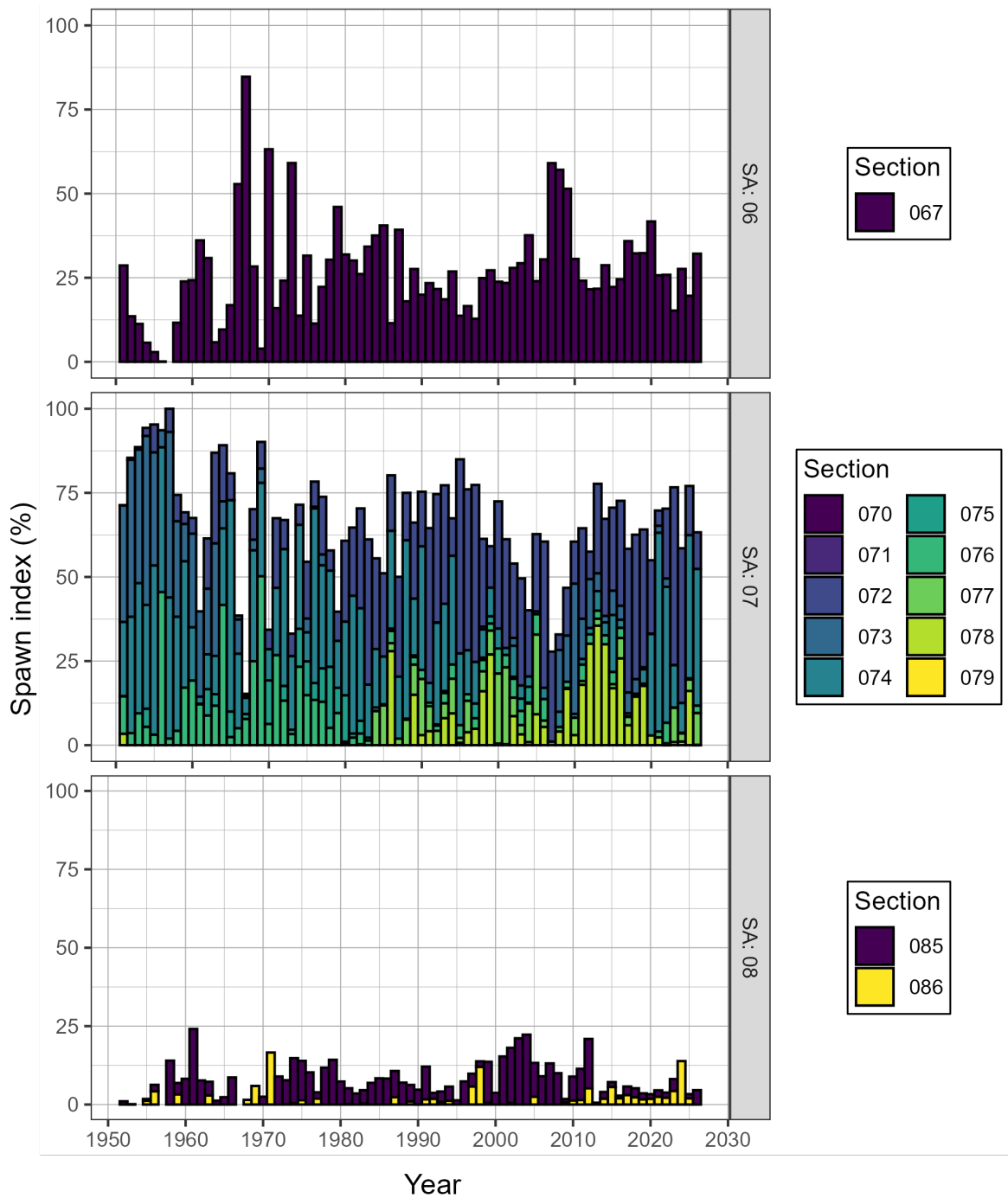


Figure 29. Time series of percent of spawn index by Statistical Area (SA) and Section for Pacific Herring from 1951 to 2026 in the Central Coast major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Legend: ‘06&07’ is Statistical Areas 06 and 07; and ‘08’ is Statistical Area 08.

Figure 30. Animation of Pacific Herring spawn index in metric tonnes (t) by Location from 1951 to 2026 in the Central Coast major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The ‘spawn index’ is not scaled by the spawn survey scaling parameter, q . Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km). View the animation: [download the report](#), open with Adobe, enable Java, and click “play”.